

Cross-cohort SLE transfer reveals a pseudobulk advantage over frozen Geneformer donor representations

Systemic lupus erythematosus single-cell RNA-sequencing study

Abstract

Patient-level molecular classifiers derived from single-cell data must retain disease information when the hospital, assay, and patient population change. Yet most representation comparisons are performed within one dataset. We compared mean-pooled embeddings from frozen Geneformer with highly variable gene (HVG) and PCA pseudobulk for donor-level systemic lupus erythematosus (SLE) discrimination in three public cohorts: GSE135779 (44 donors), GSE174188 CD4-positive cells (261 donors), and GSE285773 CD4-positive cells (26 donors). The evaluation combined 20 repeated donor-level cross-validations, reciprocal CD4-positive cohort transfer with source-fitted preprocessing, source-size learning curves, source-only representation fusion, a direct stress test of distribution-preserving Geneformer pooling, biological-program attribution, and cell-budget sensitivity.

HVG pseudobulk achieved a higher mean AUC than Geneformer in all three internal analyses (0.966 versus 0.937, 0.988 versus 0.974, and 0.936 versus 0.925). Both pseudobulk representations also achieved higher AUCs in each transfer direction. With GSE174188 as the 261-donor target, HVG and PCA pseudobulk reached 0.919 and 0.926, compared with 0.884 for Geneformer (paired Benjamini-Hochberg-adjusted DeLong $p=0.015$ and 0.000603). Source-only early fusion did not improve on pseudobulk. Seven alternatives that retained coordinate dispersion, quantiles, or low-dimensional moments produced direction-specific gains, but none passed a prespecified bidirectional continuation rule. The pseudobulk advantage expanded as the small source cohort grew. Geneformer donor scores were nearly unchanged from 1,000 to all 277,051 available cells in GSE285773. Removing a 15-gene interferon-associated component reduced Geneformer performance in every repeated split, although matched structured expression modules produced similar decreases.

These results establish expression pseudobulk as the stronger and more data-efficient transfer representation in the evaluated SLE cohorts. The study adds a strict external-transfer benchmark, a direct incremental-information test, a bidirectional pooling stress test, disease-program attribution, and empirical resource plateaus. Its conclusions apply to frozen cell embeddings with fixed aggregation and define the external standard that fine-tuned or learned patient-representation models must exceed.

Keywords: systemic lupus erythematosus; single-cell RNA sequencing; donor-level representation; Geneformer; cross-cohort transfer; pseudobulk

Introduction

Systemic lupus erythematosus (SLE) is a heterogeneous autoimmune disease, and a blood type I interferon signature is a well-recognised disease marker (Baechler et al., 2003; Bennett et al., 2003; Rönnblom and Leonard, 2019). Single-cell RNA sequencing can resolve the immune-cell states that

underlie this heterogeneity. Yet diagnosis, treatment response, and prognosis are assigned to patients, whereas the measurement is made in individual cells. This patient-cell mismatch is the central representation problem for clinical single-cell studies.

Cells from the same donor are correlated subsamples of one biological replicate (Zimmerman et al., 2021). A donor-level representation is therefore needed whenever single-cell measurements are used to study a clinical phenotype. Pseudobulk aggregation is an established way to preserve the donor as the unit of inference (Crowell et al., 2020; Squair et al., 2021).

Many donor-level representations have been proposed. Supercell analyses average cells to reduce measurement variability (Candia et al., 2013). CloudPred learns a phenotype model from variable-size cell sets and has been applied to lupus classification (He et al., 2022). scFeatures combines composition, cell-type-specific expression, pathways, and interactions (Cao et al., 2022). Other approaches model the distribution of cellular states or multicellular programs (Chen et al., 2020; Ramirez Flores et al., 2023; Joodaki et al., 2024). Together, these approaches focus attention on the donor representation that remains effective in an independent cohort.

A pretrained cell model offers a simple alternative to training a new donor-level architecture. Geneformer encodes cell transcriptomes with a pretrained transformer and supports downstream biological prediction (Theodoris et al., 2023). scGPT and scFoundation illustrate the rapid expansion of pretrained single-cell representations (Cui et al., 2024; Hao et al., 2024). Here, we average the frozen Geneformer embeddings of a donor's cells. This permutation-invariant summary is simple, inexpensive, and directly comparable across donors. Its value, however, must be demonstrated against expression pseudobulk, which is a strong donor-level baseline. We also report scVI donor means as a secondary within-cohort comparator (Lopez et al., 2018).

The decisive test is cross-cohort transfer. A molecular classifier has clinical meaning only if it retains discrimination when hospital, sequencing chemistry, and patient population change; same-distribution accuracy alone cannot establish this. We therefore make reciprocal transfer between adult and pediatric CD4-positive SLE cohorts the primary evaluation target. These cohorts differ along multiple co-occurring biological and technical axes, so age is one component of the shift rather than its sole explanation. Integration benchmarks show that reducing batch effects can also remove biological variation (Luecken et al., 2022). We therefore evaluate a source-only transfer setting: feature selection, scaling, PCA, and classifier tuning are learned in the training cohort and then applied unchanged to the target cohort. This design asks a direct question: which donor representation best preserves SLE discrimination after cohort shift?

We compare mean-pooled Geneformer embeddings with HVG and PCA pseudobulk in three public SLE cohorts. The study contributes four linked advances. First, bidirectional external transfer with source-

fitted preprocessing and paired, multiplicity-adjusted inference reveals a representation ranking that internal accuracy alone does not establish. Second, source-size learning curves and source-only fusion determine when the pseudobulk advantage appears and whether Geneformer adds transferable information beyond expression. Third, a direct reciprocal transfer experiment tests whether retaining cell-embedding dispersion, quantiles, or low-dimensional moments repairs mean pooling. Fourth, interferon residualisation, matched modules, nested cell budgets, and a full-cell probe connect discrimination to a recognised SLE axis while defining practical resource plateaus. Together, these analyses turn donor representation choice into a testable external-validation problem.

Recent work makes this distinction especially timely. PaSCient learns multicellular patient representations at atlas scale, and a broad cancer benchmark shows that aggregation strategy and simple expression baselines can materially alter patient-level foundation-model rankings (Liu et al., 2026; Elmarakeby et al., 2025). CloudPred and related methods have already established that learned cell-set models can predict lupus phenotypes (He et al., 2022). What remained unresolved was whether a frozen cell foundation model retains an advantage under label-blind, source-fitted transfer between independent autoimmune cohorts, and whether any embedding signal is incremental to pseudobulk. This study supplies that external test and links its outcome to source sample size, biological programs, and compute budgets.

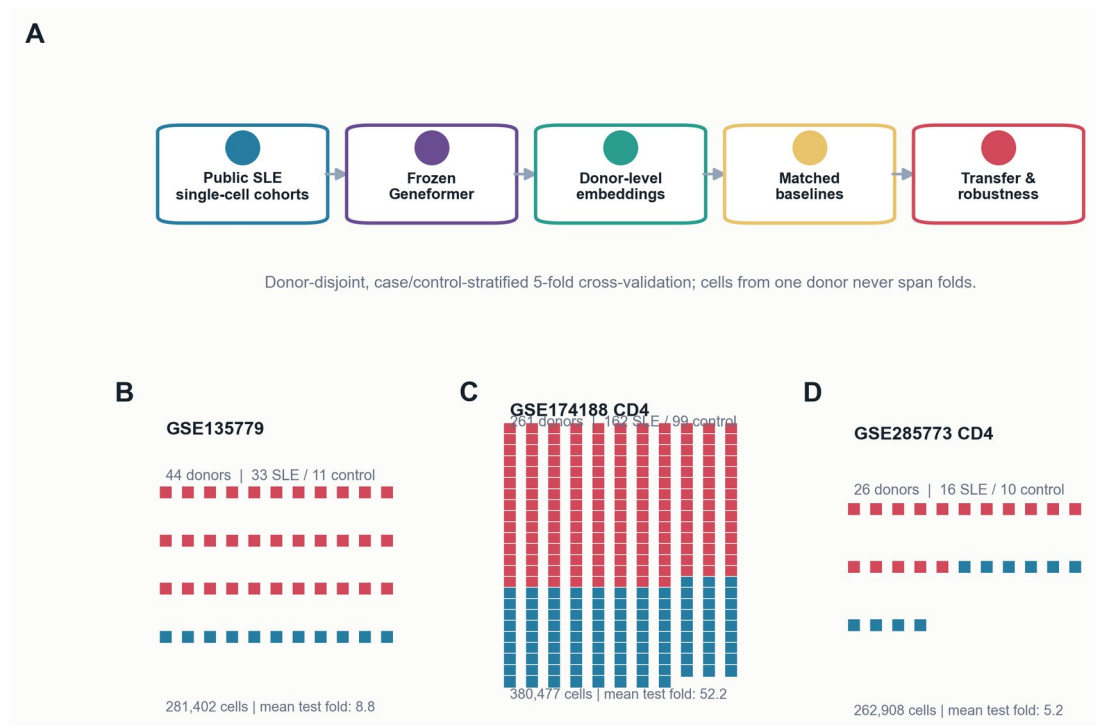


Figure 1. Donor-level representation study design. Cell-level transcriptomes were converted either to mean-pooled Geneformer embeddings or donor-level expression representations, then evaluated with donor-level internal folds and external CD4-positive cohort transfer. Cohort inventories show the number of donors contributing to each analysis.

Materials and Methods

Cohorts and study design

We reanalysed three public SLE scRNA-seq datasets: GSE135779 (44 donors: 33 cases and 11 controls), GSE174188_CD4 (261 donors: 162 cases and 99 controls), and GSE285773_CD4 (26 donors: 16 cases and 10 controls). GSE135779 profiled SLE heterogeneity (Nehar-Belaid et al., 2020), GSE174188 used multiplexed single-cell profiling (Perez et al., 2022), and GSE285773 profiled pediatric SLE CD4-positive T cells (Balasubramanian et al., 2025). We used the two CD4-positive cohorts for bidirectional transfer. Donors were the statistical unit in every supervised analysis; all cells from a donor remained in the same fold (Zimmerman et al., 2021; Crowell et al., 2020).

The primary representation used Geneformer V2-316M, a cap of 1000 sampled cells per donor, and maximum sequence length 4096. We embedded each cell with the frozen model, extracted the CLS-token vector, and averaged vectors within donor. The checkpoint was ctheodoris/Geneformer revision 04c2b2e84da7c0f385c3f9ad8f3ec24bab6650e5. The main comparators were mean-HVG pseudobulk and PCA pseudobulk. For internal evaluation, we selected the 4,000 most variable genes and fitted PCA within each outer training fold, then applied the fitted transformation to held-out donors. We used balanced liblinear logistic regression ($C=1.0$, $\text{max_iter}=5000$) for all internal comparisons. Dataset-specific scVI donor means appear in the fixed-split within-cohort reference panels.

Internal evaluation and statistical inference

The primary internal analysis repeated stratified five-fold donor-level cross-validation 20 times with split seeds 20260801-20260820. Every method used the same donor assignments in each repeat. We report repeat-level ROC-AUC means, standard deviations, and 2.5th-97.5th percentiles to show sensitivity to donor splits. The original five-fold split (`StratifiedKFold(n_splits=5, shuffle=True, random_state=1)`) was used for the cell-budget and interferon analyses. PR-AUC (Saito and Rehmsmeier, 2015), Brier score (Brier, 1950), and 10-bin expected calibration error (ECE) were secondary metrics. ECE used 10 equal-width probability bins (Guo et al., 2017; Van Calster et al., 2019).

For fixed-split and transfer predictions, we calculated ROC-AUC intervals from 5,000 case/control-stratified target-donor bootstrap resamples using the percentile method (Efron and Tibshirani, 1993). Transfer intervals are conditional on the single fitted source model and therefore quantify target-donor sampling uncertainty, not source-model refitting uncertainty. Paired DeLong tests compared representations on the same target donors (DeLong et al., 1988). We adjusted the four primary transfer comparisons with the Benjamini-Hochberg procedure (Benjamini and Hochberg, 1995). Repeated internal results are summarized as performance distributions because repeats reuse the same donors.

Cross-cohort transfer and cell-budget sensitivity

For each CD4-positive transfer direction, we trained logistic regression in the source cohort and evaluated it once in the independent target cohort. Pseudobulk inputs were donor-specific $\log_{10}(1e6 * \text{summed raw counts} / \text{donor library size})$ matrices. Target-unlabelled feature-schema harmonisation used the exact intersection of whitespace-trimmed, unique retained identifiers; no target expression values or labels entered feature ranking. The source cohort then selected the 4,000 greatest population-variance genes, fitted scaling and up to 30 PCs (24 when the 26-donor cohort was the source), and selected C from 10^{-4} to 10^4 by stratified cross-validation. We applied these transformations and the classifier unchanged to the target cohort. Geneformer vectors were scaled with source-cohort statistics. Target-centered and target-z-scored analyses are reported separately as transductive sensitivity analyses.

Because the archived primary transfer selected regularisation with method-specific source-internal folds, we performed an additional label-blind sensitivity in which all three representations shared the same source-internal donor folds. No target labels, target expression summaries, or target-fitted transformations entered this sensitivity. It was used to test whether method ordering depended on the internal fold assignment and was not substituted post hoc for the prespecified primary estimates.

To test incremental information, we concatenated source-scaled Geneformer with either source-HVG or source-PCA pseudobulk, rescaled the combined block on source donors, and selected regularisation by the same source-only cross-validation. The target cohort was transformed with source-fitted parameters. We compared each hybrid with its pseudobulk component and with Geneformer on identical target donors; eight paired DeLong tests were Benjamini-Hochberg adjusted.

We evaluated Geneformer embeddings generated with fixed sampling seeds and caps of 200, 500, and 1000 cells per donor. The matched 500-versus-1000 analysis used the same donors, labels, and folds at both caps. In GSE285773, an additional all-available-cell run used 277,051 cells, compared with 26,000 cells at the 1,000-cell cap. We report bootstrap intervals for AUC differences, paired DeLong tests adjusted across the three cohorts, donor-score Spearman correlations, and embedding similarity. The 200-to-1000-cell transfer results are shown as supplementary sensitivity analyses because they used a historical transfer workflow.

For the GSE285773-to-GSE174188 learning curve, we drew 40 stratified source-donor subsets without replacement at $n=10, 14, 18,$ and 22 . Feature selection, scaling, PCA, and regularisation were refitted in each subset, while the 261-donor target remained fixed and unlabeled until evaluation. The $n=26$ endpoint is the single prespecified full-source result rather than 40 duplicate full-cohort resamples.

Distribution-preserving Geneformer pooling sensitivity

We directly tested whether ordinary coordinate averaging discarded transferable information present in frozen Geneformer cell embeddings. This prespecified cap500 experiment used Geneformer V2-316M revision 04c2b2e84da7c0f385c3f9ad8f3ec24bab6650e5, sampling seed 1, maximum sequence length 4096, and identical 500-cell donor samples in both reciprocal CD4-positive transfer directions. The cap500 cell embeddings reproduced the archived donor means before evaluation (median donor cosine greater than 0.999999996 in both cohorts). We compared coordinate mean with coordinate median, 10% coordinate-wise trimmed mean, mean plus standard deviation, source-fitted cell-PCA32 mean plus 10th/50th/90th quantiles, source-fitted cell-PCA16 mean plus shrunk covariance moments, and source-PCA8 or source-PCA16 donor means. Cell-state-stratified means were not evaluated because GSE285773 lacked a harmonised CD4 subtype annotation.

All operators and downstream transformations were fitted in the source cohort. Balanced logistic regression selected C from 10^{-4} to 10^4 by five-fold stratified source-donor cross-validation, then scored the independent target donors without target-label tuning. Each candidate was compared with ordinary mean pooling on the same target donors using 5,000 case/control-stratified paired bootstrap resamples and paired DeLong tests; the 14 candidate-by-direction DeLong tests were Benjamini-Hochberg adjusted. The prespecified cap1000 continuation rule required an AUC increase of at least 0.010 in the 261-donor target, no AUC loss in the 26-donor target, and no candidate-minus-mean deterioration greater than 0.02 in Brier score or 0.05 in 10-bin ECE in either direction. No candidate met this rule, so the sequence stopped at cap500.

Interferon-conditioned representation sensitivity

We analysed the interferon association in GSE174188 CD4-positive alpha-beta T cells. The interferon-stimulated-gene (ISG) score was the mean donor-pseudobulk log_{1p} CPM expression of a fixed 15-gene panel: ISG15, IFI6, MX1, OAS1, OAS2, OAS3, IFIT1, IFIT3, IFI44, IFI44L, STAT1, RSAD2, IFITM1, IFITM3, and HERC5 (Baechler et al., 2003; Bennett et al., 2003; Rönnblom and Leonard, 2019).

Within each training fold, we scaled the ISG score and fitted one linear model from the score to each of the 1,152 Geneformer coordinates. We then removed the fitted ISG-associated component from test-donor embeddings. The fixed-fold analysis compared Geneformer, ISG-only, and ISG-residualized classifiers. The 1,000-cell analysis repeated 20 stratified donor splits and added an ISG-plus-residual classifier.

To test the specificity of the association, we generated 500 non-ISG modules for each of the first five repeated splits. Each module matched the ISG panel in donor expression, variance, score variance, and

mean pairwise correlation. ISG and random-module scores used all available CD4 cells per donor because exact sampled-cell identifiers were unavailable.

Supporting analyses, software, and reproducibility

Supplementary Figure S1 reports the observed, incomplete model-size and maximum-length configurations. A separate scGPT 0.2.4 analysis evaluated 237 GSE174188 donors with ten nested within-donor samples at 25, 50, 100, 200, and 500 cells per donor. It characterizes low-cell-count performance and sampling variability. The archived whole-human checkpoint used scGPT 0.2.4 with Python 3.10 and PyTorch 2.5.1+cu124; checkpoint, vocabulary, arguments, and canonical embedding hashes are included in the provenance record.

An additional archived sensitivity used 261 GSE174188 donors, 500 cells per donor, and 30 matched donor-disjoint five-fold repeats to compare scGPT donor means, multiscale kernel mean embeddings of scGPT cell vectors, and donor-expression PCA. Because the same donors recur across splits, paired repeat differences are descriptive and no repeat-level p values are reported.

We also characterized the geometry of the cap500 donor-mean Geneformer matrices without using outcome labels for dimensionality reduction. PCA coordinates, entropy and participation effective ranks, and case-control centroid separation were calculated separately in the two reciprocal-transfer cohorts. Cell-level spectra were estimated from deterministic samples of 20,000 cap500 cell embeddings per cohort. Cap500 and cap1000 spaces were compared with raw and globally centered donor-wise cosine similarity, coordinate RMSE, and correlation between donor-distance matrices.

We additionally recovered a broad fixed-split aggregation atlas from the immutable scGPT analysis snapshot. It evaluated 22 distinct donor representations on the same 261 GSE174188 donors and the same matched 500-cell, five-fold donor split. The atlas covered expression pseudobulk, frozen scGPT mean pooling, moments and quantiles, kernel mean embeddings, prototype summaries, Deep Sets, attention and top-k multiple-instance learning, a Set Transformer, and exploratory donor-set operators. We retained exact aliases and incomplete methods in the provenance table but removed exact duplicate aliases from the plotted ranking. Paired donor bootstrap intervals compare each completed method with scGPT mean pooling. This single-cohort atlas is exploratory: it expands the tested aggregation space but does not replace the repeated internal or external-transfer analyses.

Three archived control branches were also incorporated. First, 1,000 label permutations were completed for scGPT mean, expression PCA, and multiscale KME in GSE174188; empirical p values used $(1 + \text{number of null AUCs at least as large as observed}) / (1 + 1000)$. Second, an accepted covariate sensitivity fitted training-fold residualization for cells per donor, mean UMI per cell, mean detected genes per cell, age, and sex before refitting donor classifiers. Third, descriptive kernel diagnostics recorded training-fold

bandwidth, effective rank, and prediction agreement for 25-500 cells per donor. Historical Geneformer controls included 1,000 label permutations in each cohort, total-cell-count-only prediction, univariate Geneformer-PC associations with cell yield, classifier ablations, model-size/sequence-length probes, and a whole-PBMC cell-composition baseline. These branches use fixed splits and heterogeneous estimands; they are treated as supporting diagnostics.

The final cap500 Geneformer pooling experiment ran under Python 3.12.3, PyTorch 2.8.0+cu128, Transformers 4.53.2, scikit-learn 1.7.0, and CUDA 12.8. The frozen model file SHA256 was 965ceceea81953d362081ef3843560a0e4fef88d396c28017881f1e94b1246f3. The local statistical and visualization reanalysis environment is separately locked and includes scanpy 1.12.2 and scikit-learn 1.9.0. The exact historical scanpy, scvi-tools, and scikit-learn pins used to generate the original fixed-split scVI reference were not retained and are not reconstructed retrospectively. The reproducibility package includes scripts, donor-level predictions, metrics, fold assignments, input and embedding hashes, random seeds, the Geneformer revision and model hash, the scGPT checkpoint and configuration records, and environment locks for the retained computational branches.

Results

Pseudobulk outperformed Geneformer in repeated within-cohort cross-validation

Mean-pooled Geneformer embeddings discriminated SLE cases from controls in all three cohorts. Across 20 repeated five-fold donor splits, mean AUC was 0.937 in GSE135779, 0.974 in GSE174188 CD4-positive cells, and 0.925 in GSE285773 CD4-positive cells (Table 1). The smallest cohort, GSE285773, showed the widest split-to-split variation (2.5th-97.5th percentile, 0.840-0.975).

HVG pseudobulk achieved higher mean AUCs than Geneformer in GSE135779 (0.966), GSE174188 (0.988), and GSE285773 (0.936). PCA pseudobulk achieved mean AUCs of 0.954, 0.978, and 0.890, respectively. The ranking was consistent with the fixed-split reference curves (Supplementary Figure S4). These results show that Geneformer captures donor-level disease information and that HVG pseudobulk yielded the strongest within-cohort performance.

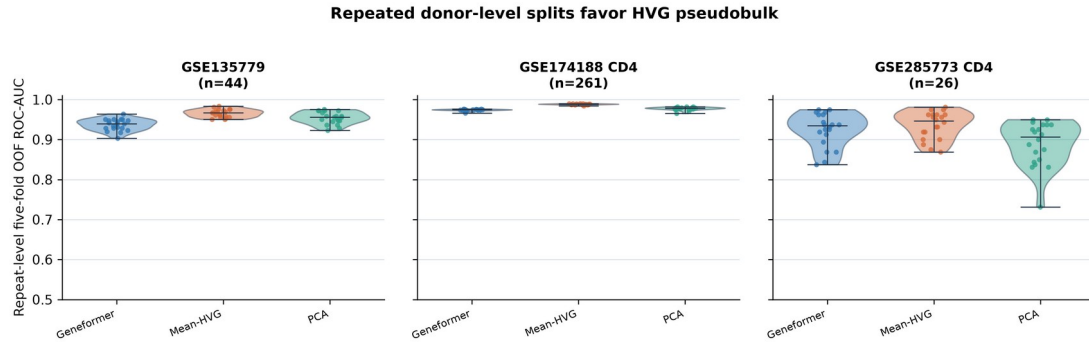


Figure 2. Repeated within-cohort benchmark. Each point is the donor-level out-of-fold ROC-AUC from one of 20 repeated stratified five-fold splits. All methods used the same donor assignments within each repeat. HVG selection and PCA fitting used training donors only. Violin shapes show the distribution across repeats.

Supplementary Figure S14 displays the paired repeat-level AUC differences that underlie these distributions. Supplementary Figure S15 provides label-blind donor-level projections of the three representations, and Supplementary Figure S16 brings the repeated and external AUC estimates into one forest plot.

Pseudobulk outperformed Geneformer in cross-cohort transfer

When GSE174188 was the source and GSE285773 was the target, Geneformer achieved AUC 0.900. HVG and PCA pseudobulk achieved higher AUCs of 0.944 and 0.981. The paired BH-adjusted p values were 0.491 and 0.106.

The reverse direction provided a larger test set. When GSE285773 was the source and GSE174188 was the 261-donor target, Geneformer achieved AUC 0.884 (95% CI 0.843-0.920). HVG and PCA pseudobulk achieved AUCs of 0.919 (0.884-0.946) and 0.926 (0.895-0.953). Paired DeLong tests on the same target donors favored both pseudobulk representations after adjustment (BH $p=0.015$ and 0.000603). Geneformer transferred across cohorts; pseudobulk was the stronger representation in both directions. The reciprocal transfer spans cohorts with multiple co-occurring shifts, including pediatric and adult recruitment, and shows that the disease signal generalises across the evaluated populations while remaining better preserved by pseudobulk.

A shared-fold source-internal sensitivity analysis preserved this external ordering. In the 261-donor target, Geneformer, HVG pseudobulk, and PCA pseudobulk achieved AUCs of 0.890, 0.916, and 0.916; both paired pseudobulk comparisons remained significant after BH correction ($q=0.029$). The larger-target pseudobulk advantage therefore was not created by method-specific source-fold assignments (Supplementary Figure S29).

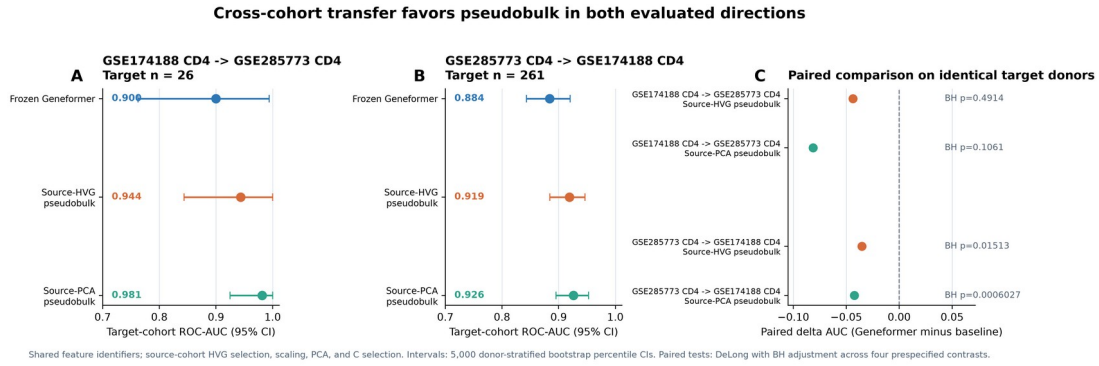


Figure 3. Cross-cohort transfer favors pseudobulk. Gene identifiers were aligned across cohorts, and feature selection, scaling, PCA, and regularization were fitted in the source cohort. Points show target-cohort ROC-AUC with 5,000-resample donor-stratified 95% intervals. The larger GSE174188 target direction provides paired evidence that both pseudobulk representations outperformed Geneformer.

Target-donor score distributions and the full ROC-AUC, PR-AUC, and Brier-score matrix are shown in Supplementary Figures S9-S10. Supplementary Figure S17 maps the paired target-donor probabilities assigned by Geneformer and pseudobulk, showing where the representations agree and diverge. Supplementary Figure S22 presents descriptive 10-bin reliability curves for the same strict transfer predictions.

Frozen Geneformer did not add transfer performance beyond pseudobulk under early fusion

Source-only fusion directly tested whether the frozen embedding contained transferable information that pseudobulk lacked. In the GSE174188-to-GSE285773 direction, Geneformer-plus-HVG and Geneformer-plus-PCA achieved AUCs of 0.887 and 0.900, below HVG (0.944) and PCA pseudobulk (0.981). In the reverse direction, the hybrids achieved 0.906 and 0.890, below the corresponding pseudobulk AUCs of 0.919 and 0.926. Geneformer-plus-HVG improved on Geneformer alone in the 261-donor target (difference 0.022; BH-adjusted $p=0.036$) but did not improve on HVG pseudobulk (difference -0.013; BH-adjusted $p=0.253$). Geneformer-plus-PCA was lower than PCA pseudobulk in the same direction (difference -0.036; BH-adjusted $p=0.0083$). Thus, frozen Geneformer carried transferable disease signal, but simple early fusion did not convert it into incremental performance beyond pseudobulk.

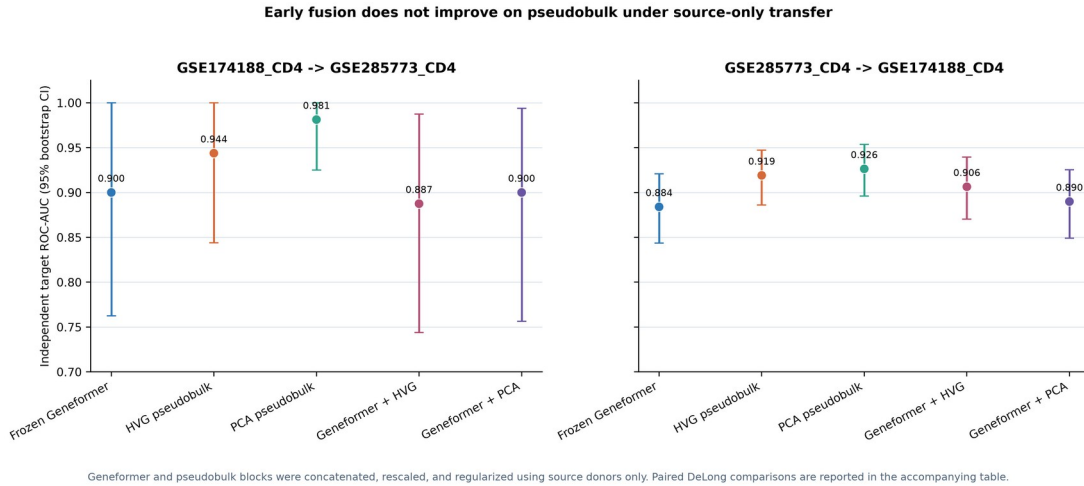


Figure 4. Source-only fusion tests incremental transfer information. Geneformer vectors were concatenated with HVG or PCA pseudobulk, then rescaled and regularised using source donors only. Points show independent-target ROC-AUC with target-donor bootstrap intervals. Neither hybrid exceeded its corresponding pseudobulk representation.

Distribution-preserving Geneformer pooling produced directional, not general, gains

The direct cap500 experiment tested seven alternatives to ordinary Geneformer coordinate averaging under the same reciprocal source-only transfer design (Figure 5; Table 5). Ordinary mean pooling reached AUC 0.863 when GSE285773 was the 26-donor target and 0.884 when GSE174188 was the 261-donor target. Coordinate median increased AUC by 0.056 in the smaller target but reduced AUC by 0.026 in the larger target (BH-adjusted DeLong $q=0.000559$ for the reduction). Source-PCA8 donor mean showed an even wider reversal, increasing AUC by 0.081 in the smaller target while decreasing it by 0.050 in the larger target ($q=0.000838$ for the reduction).

The 10% trimmed mean was the closest candidate to a stable gain. It increased AUC in the 261-donor target by 0.00761 (paired bootstrap 95% interval, 0.00025 to 0.01571) but decreased AUC in the 26-donor target by 0.00625. Its paired DeLong p value in the larger target was 0.041 before multiplicity adjustment and $q=0.096$ across all 14 exploratory comparisons. It therefore fell below the prespecified 0.010 continuation threshold and did not support a cap1000 run. Mean plus standard deviation was effectively unchanged in the larger target (difference 0.0010), while quantile and covariance summaries were directionally unstable. Several small-target models also had Brier score and ECE near 0.62 despite useful ranking. Fixed distribution summaries can therefore change which donors are ranked correctly without producing a robust bidirectional representation or calibrated clinical probabilities.

Distribution-preserving Geneformer pooling produced directional, not general, transfer gains

Frozen V2-316M cell embeddings; 500 cells per donor; source-fitted preprocessing; reciprocal independent-cohort evaluation

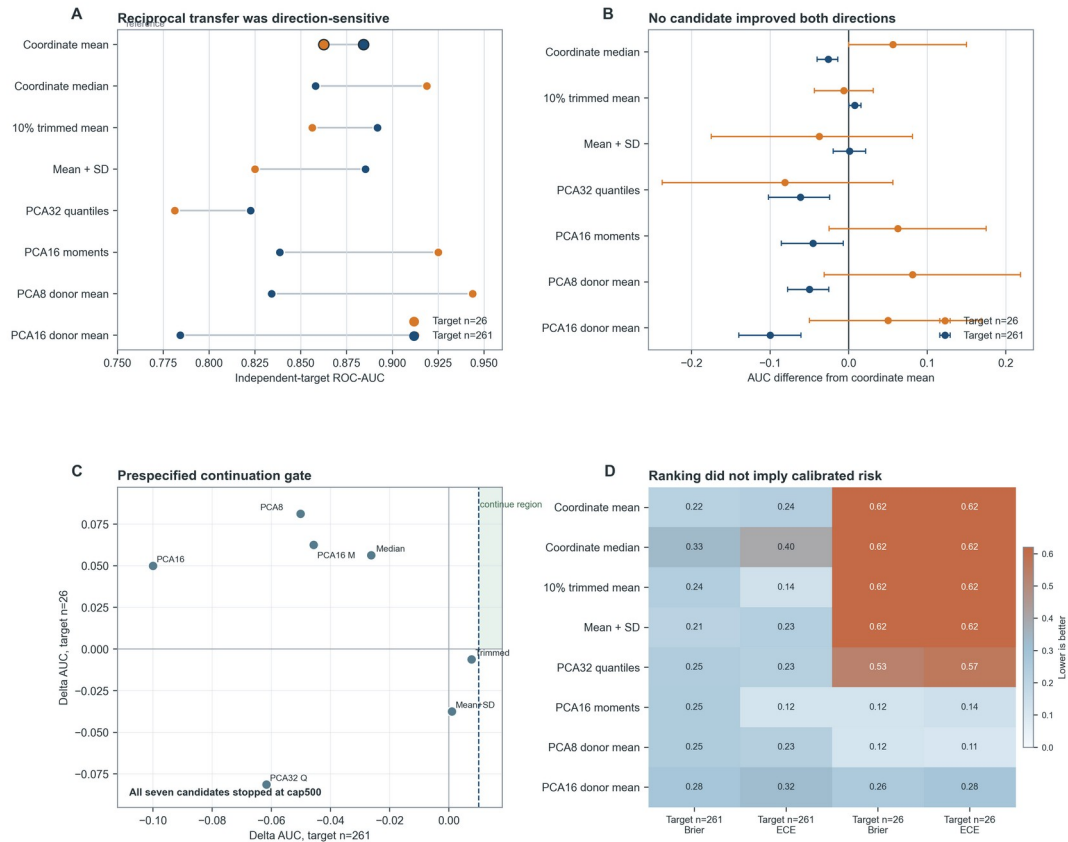


Figure 5. Distribution-preserving Geneformer pooling under reciprocal source-only transfer. Panel A shows independent-target ROC-AUC for ordinary mean pooling and seven fixed alternatives at 500 cells per donor. Panel B gives candidate-minus-mean AUC differences with 5,000-resample paired stratified bootstrap intervals. Panel C maps each candidate against the prespecified continuation thresholds; none entered the continuation region. Panel D reports Brier score and 10-bin ECE in both target cohorts. All pooling operators, scaling, dimensionality reduction, and regularisation were fitted in the source cohort.

Resource scaling and aggregation reached clear performance plateaus

In the matched Geneformer analysis, increasing the cap from 500 to 1000 cells changed AUC by 0.0248 in GSE135779, 0.0006 in GSE174188 CD4, and 0.0063 in GSE285773 CD4. The corresponding bootstrap intervals were 0.0028-0.0551, -0.0019-0.0036, and -0.0313-0.0500. BH-adjusted paired p values were 0.123, 0.692, and 0.692. Donor-level scores remained highly correlated at the two caps (Spearman rho 0.989, 0.999, and 0.995).

The independent scGPT analysis showed the complementary low-cell-count trend in GSE174188 CD4. On 237 donors, scGPT donor-mean AUC increased from 0.937 at 25 cells to 0.963 at 100 cells and 0.974 at 500 cells across ten nested samples. Sampling variability decreased from 0.010 to 0.002. Expression PCA improved from 0.960 to 0.983 and remained the best-performing representation at every cap (Supplementary Figure S3).

Two archived analyses extended the upper and algorithmic bounds. In GSE285773, increasing Geneformer input from 26,000 cells at the 1,000-cell cap to all 277,051 available cells left AUC unchanged at 0.931 and preserved donor ranking (Spearman $\rho=0.9993$; median embedding cosine=0.999990). In the 261-donor, 30-repeat GSE174188 sensitivity, scGPT mean pooling achieved mean AUC 0.9773, multiscale kernel mean embedding achieved 0.9747, and expression PCA achieved 0.9858. Expression PCA exceeded scGPT mean in 30/30 matched repeats, whereas kernel mean embedding exceeded scGPT mean in 6/30. More cells and a distribution-aware fixed pooling operator therefore did not automatically improve the donor representation (Supplementary Figure S23).

Supplementary Figure S1 shows model-size and maximum-length implementation sensitivity.

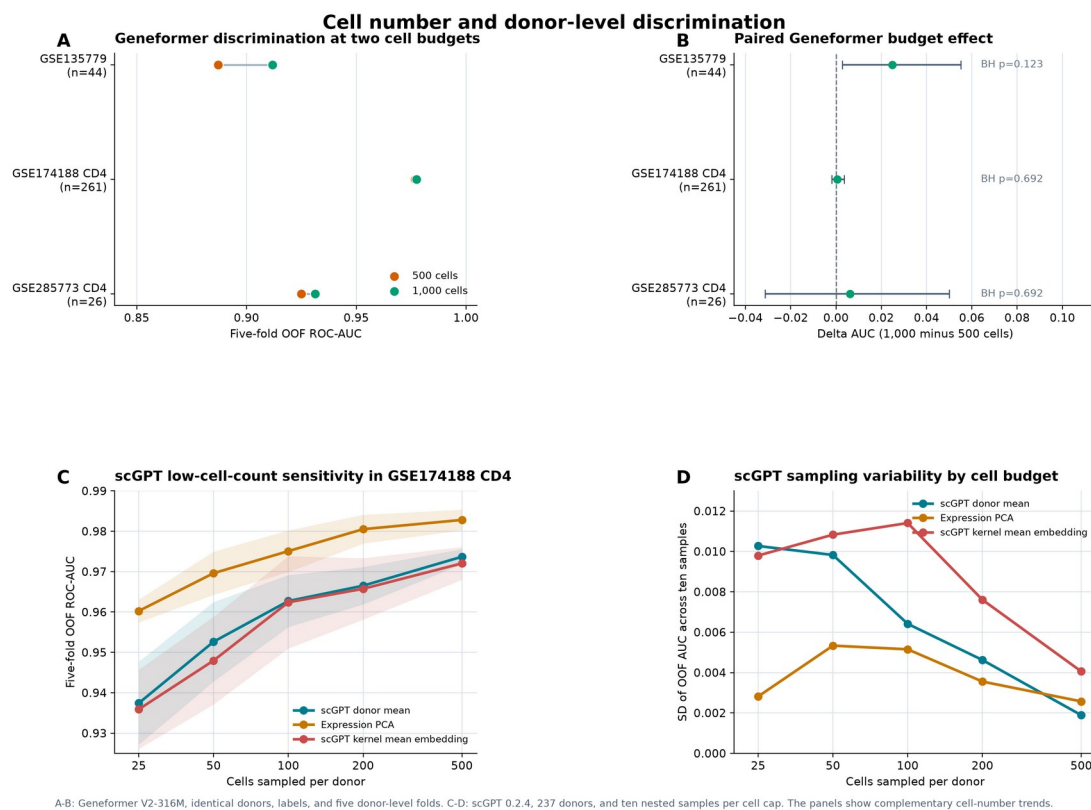


Figure 6. Cell number and donor-level discrimination. Panels A-B show the matched Geneformer V2-316M analysis at 500 and 1000 cells per donor. Panel B shows paired AUC differences with 5,000-resample bootstrap intervals and BH-adjusted paired DeLong p values. Panels C-D show mean performance and sampling variability in the independent scGPT low-cell-count analysis.

To test whether the external advantage depended on the small source cohort, we repeatedly subsampled the 26 GSE285773 donors and transferred to the fixed 261-donor GSE174188 target. At 10 source donors, mean target AUCs were closely matched (Geneformer 0.850, HVG pseudobulk 0.847, PCA pseudobulk 0.842). By 22 source donors, the corresponding means were 0.872, 0.907, and 0.897. The single prespecified full-source endpoint reached 0.884, 0.919, and 0.926. The widening separation

shows that pseudobulk converted additional source donors into external discrimination more effectively in this cohort pair (Supplementary Figure S18).

The donor-mean Geneformer geometry was markedly lower-dimensional than the sampled cell space. Entropy effective rank was 6.8 and 6.7 for the two donor-mean matrices, compared with 51.4 and 31.8 in the sampled cell spectra. Nevertheless, cap500 and cap1000 donor geometries were nearly identical: centered cosine medians were 0.998 and 0.993, and donor-distance-matrix correlations were 0.999 and 0.997. These findings distinguish a stable cell-budget plateau from evidence that coordinate averaging retains the full cell-level geometry (Supplementary Figure S30).

A broad aggregation atlas reinforced the strength of simple donor summaries

The recovered GSE174188 atlas widened the comparison from three primary representations to 22 distinct aggregation strategies under one matched fixed split (Supplementary Figure S24; Table 3). Raw pseudobulk ranked first (AUC 0.988), followed by expression PCA (0.986), mean-HVG pseudobulk (0.984), scGPT mean pooling (0.978), and multiscale KME (0.978). Attention MIL reached 0.970, top-k MIL 0.969, Deep Sets 0.960, and the Set Transformer 0.954. Their paired AUC differences from scGPT mean were -0.0084, -0.0096, -0.0184, and -0.0241, respectively; the bootstrap intervals excluded zero for Deep Sets and the Set Transformer but not attention or top-k MIL. Distribution summaries such as quantiles, tail fractions, and prototype histograms also failed to exceed mean pooling.

The associated historical three-cohort matrix contained 58 completed method records spanning expression, Geneformer, and scGPT branches (Supplementary Figure S25). Although the matrix combines different historical implementation branches and has missing method-by-cohort cells, its pattern was consistent with the primary result: raw or PCA/HVG pseudobulk occupied the top tier in the large cohort and in the 26-donor cohort, while several more elaborate aggregators were competitive in the 44-donor cohort but did not establish a stable global advantage. The maximal atlas therefore identifies learned pooling as a plausible future direction while showing that architectural complexity alone is not the missing source of transportability.

Null, covariate, and representation diagnostics bounded alternative explanations

Formal 1,000-permutation tests in GSE174188 gave empirical $p=0.000999$ for scGPT mean (observed AUC 0.978), expression PCA (0.986), and multiscale KME (0.978), confirming that the fixed-split discrimination was far beyond label-randomized performance (Supplementary Figure S26). Independent Geneformer permutation controls across the three cohorts likewise yielded empirical p values from 0.000999 to 0.001998 against the archived pipeline AUCs.

Available technical and demographic variables were themselves predictive in GSE174188 (covariates-only AUC 0.789). Training-fold residualization attenuated scGPT mean from 0.978 to 0.913, expression PCA from 0.986 to 0.885, and mean-HVG pseudobulk from 0.984 to 0.916 (Table 4). Thus, measurable donor and technical structure explained a meaningful component of all three representations, but substantial disease discrimination remained after adjustment. This is a sensitivity result rather than evidence of causal independence because medication, disease activity, ancestry, processing centre, and other covariates were unavailable or too sparse.

Kernel diagnostics further clarified why a stable distribution-aware operator need not add predictive information. Multiscale KME bandwidths remained within 7.55-7.65 across five folds and five cell caps, and its predictions were distinct from scGPT mean at every cap. However, effective rank remained only 2.6-2.8 out of up to 512 diagnostic points, indicating a strongly concentrated kernel spectrum (Supplementary Figure S27). Geneformer PC1 also showed substantial association with log total cell yield in GSE135779 ($R^2=0.27$) and GSE285773 ($R^2=0.66$), whereas the same association was weaker in GSE174188. Total cell count alone reached AUC 0.810, 0.693, and 0.319 in the three cohorts, below Geneformer in every cohort; the unmatched whole-PBMC composition baseline reached AUC 0.912 in GSE174188 (Supplementary Figure S28). These controls show that nuisance structure is present and measurable without reducing the donor-level result to a single technical proxy.

Geneformer discrimination was associated with an interferon-related expression axis

The donor ISG score separated cases from controls with AUC 0.854 (95% CI 0.809-0.897), consistent with the established SLE interferon signature (Baechler et al., 2003; Bennett et al., 2003; Rönnblom and Leonard, 2019). At 1000 cells, Geneformer AUC was 0.977 (0.960-0.991). Removing the linear ISG-associated component reduced AUC to 0.818 (0.765-0.868); the 500-cell analysis gave 0.819 (0.768-0.868). Across 20 repeated donor splits, residualization decreased AUC in every split (mean change -0.152), whereas adding the ISG score back restored discrimination. Matched random expression modules produced decreases of similar magnitude (empirical $p=0.307$ -0.337), linking Geneformer discrimination to broad structured expression (Supplementary Figures S7-S8) and connecting its discrimination to an interferon axis familiar in SLE care. The donor-by-gene heatmap in Supplementary Figure S20 makes this coordinated ISG gradient visible across the GSE174188 donor set.

Pseudobulk recovered established interferon-associated SLE genes

Pseudobulk summaries recovered established SLE-associated interferon genes, including IFI27, IFI44L, IFIT3, IFI6, RSAD2, and OAS1. These recurring expression differences provide a biological explanation for the strong performance of donor-level pseudobulk. Supplementary Figure S19 shows both the cohort-specific effect-size and significance landscape and the overlap of high-effect genes; Supplementary

Figure S21 shows that fixed-fold donor rankings were highly correlated across all four representations despite their performance differences.

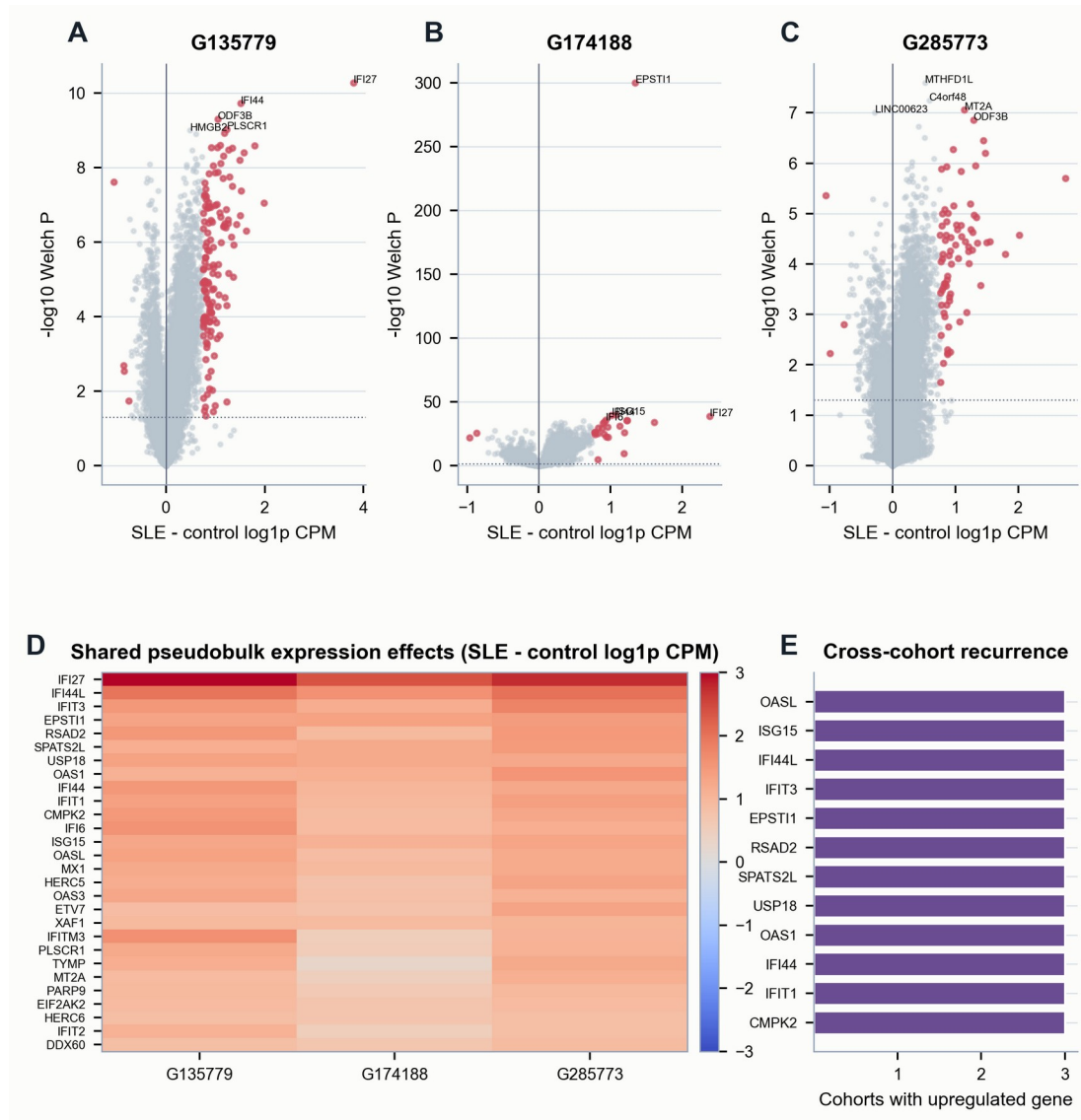


Figure 7. Pseudobulk recovers SLE-associated interferon genes. Cohort-level differential summaries identify recurring interferon-associated expression patterns across cohorts.

Supplementary Figure S12 presents the recurring high-effect genes together with complete shared-gene effect-size comparisons.

Discussion

This study establishes three empirical facts about donor representation under SLE cohort shift. First, mean-pooled frozen Geneformer embeddings retained strong disease signal, but pseudobulk was

stronger in repeated internal evaluation and in both external directions. Second, the pseudobulk advantage expanded as source donors accumulated and was not erased by early fusion with Geneformer. Third, preserving cell-embedding dispersion, quantiles, or low-dimensional moments produced striking gains in one direction only by sacrificing the other; more cells and greater fixed pooling complexity did not automatically improve donor discrimination. Representation choice is therefore a cross-centre generalisation and information-efficiency problem, not a contest in same-dataset accuracy or model scale.

Several features of the data may explain the pseudobulk advantage. SLE case-control separation in blood is strongly reflected in average donor expression, including reproducible interferon-associated genes. Pseudobulk preserves these donor-level shifts in a shared gene space. Coordinate-wise averaging of cell embeddings compresses each donor into a single vector and removes information about the distribution of cellular states, rare populations, and cell-state interactions. The reciprocal pooling experiment shows, however, that restoring generic dispersion or quantile information is not sufficient: those summaries amplified cohort-specific structure as readily as transferable disease structure. Geneformer pretraining optimizes cell-level transcriptomic representation, whereas donor-level disease discrimination depends on aggregate expression and cellular composition. These differences offer a plausible explanation for the cross-cohort advantage of pseudobulk.

Geneformer nevertheless has practical value. It achieved high AUC within every cohort and transferred above chance in both directions using its original pretrained weights. Mean pooling is inexpensive and provides a common feature space for heterogeneous cell collections. The fusion result clarifies its role: in these cohorts, the embedding was a competent standalone representation but did not supply stable incremental transfer information beyond pseudobulk. The resulting selection rule is concrete: benchmark frozen embeddings against source-fitted pseudobulk on an independent cohort, and require a hybrid or learned representation to demonstrate added external performance rather than assume complementarity.

The resource and interferon analyses add two further insights. Performance and sampling stability improved rapidly between 25 and 100 cells, changed little from 500 to 1,000 cells, and were effectively unchanged when the small cohort expanded from 26,000 sampled cells to all 277,051 available cells. The cap500 Geneformer experiment and the matched 30-repeat scGPT analysis independently show that a more elaborate fixed distributional summary is not sufficient by itself. The interferon analysis showed that Geneformer discrimination is strongly associated with a familiar SLE program: removing the linear ISG-associated component consistently reduced AUC, and adding the score back restored performance. The matched-module result places that finding in context: IFN is a biologically recognisable instance of a broader structured expression axis, not a uniquely isolated causal explanation. Relating predictive

performance to this established disease program improves biological interpretability without overstating mechanism.

Learned patient representations now provide the most important next comparison. PaSCient demonstrates that large-scale multicellular pretraining can improve patient classification and interpretation (Liu et al., 2026), while cancer benchmarking shows that aggregation and simple baselines remain decisive (Elmarakeby et al., 2025). The direct Geneformer pooling experiment and the archived KME result sharpen this agenda: preserving more of a cellular distribution is not enough unless the representation learns disease-relevant structure that transfers in both directions. Attention-based multiple-instance learning, Set Transformers, PaSCient-like pretraining, PhEMD, and PILOT should therefore be compared with pseudobulk under the same source-only cohort-transfer protocol (Ilse et al., 2018; Lee et al., 2019; Chen et al., 2020; Joodaki et al., 2024).

The maximal historical atlas strengthens this benchmark in two ways. It shows empirically that multiple distributional and learned set operators can remain below simple pseudobulk even when all methods receive the same donor split, and it documents measurable technical attenuation rather than assuming that foundation-model embeddings are nuisance-free. These results sharpen the paper's positive increment: external transfer, incremental fusion, source-size scaling, program attribution, and a broad pooling stress test now converge on the same method-selection principle.

Limitations and scope

The smaller transfer target contained 26 donors, yielding wide confidence intervals and limited power for paired comparisons; the nonsignificant primary transfer direction should be interpreted accordingly. The external analysis covers two CD4-positive cohorts and cannot separate age from the other biological and technical differences between them. The primary scope is frozen Geneformer with fixed pooling; the early-fusion and cap500 pooling results do not exclude gains from fine-tuning, supervised set learning, or cell-type-aware pooling. The pooling operators were evaluated at one sampling seed and stopped at cap500 under a prespecified gate. Repeated cross-validation characterizes split sensitivity within the available donors, and transfer bootstrap intervals are conditional on one fitted source model. The scGPT analyses use distinct archived estimands and are resource or aggregation sensitivities rather than matched cross-model transfer comparisons. The interferon analysis was performed in one cohort, used all available CD4 cells for module scores, and quantifies linear association rather than causation or IFN specificity. Public metadata did not support harmonized adjustment for medication, disease activity, recruitment centre, and all technical covariates. The observed model-size and sequence-length grid is incomplete and remains descriptive. The low-rank analysis is also descriptive and does not establish that spectral contraction itself causes the transfer gap.

Conclusion

Pseudobulk outperformed mean-pooled frozen Geneformer embeddings for donor-level SLE discrimination across internal resampling and reciprocal cohort transfer, and neither early fusion nor distribution-preserving fixed pooling produced a bidirectionally stable improvement. The clinical implication is direct: a patient-level molecular representation should be selected by its ability to preserve disease ranking across centres and populations, not by model scale or within-cohort accuracy alone. Source-size curves, biological-program attribution, reciprocal pooling tests, and resource plateaus make this benchmark reusable. For frozen cell embeddings with fixed aggregation, pseudobulk is the reference to beat; learned patient representations should demonstrate incremental value under the same external protocol.

Tables

Table 1. Repeated donor-level cross-validation. Values are mean ROC-AUC across 20 repeated stratified five-fold analyses. Intervals show the 2.5th-97.5th percentiles across repeats. Mean-HVG selection and PCA fitting used outer-training donors only.

Dataset	Frozen Geneformer mean AUC [P2.5-P97.5]	Mean-HVG pseudobulk mean AUC [P2.5-P97.5]	PCA pseudobulk mean AUC [P2.5-P97.5]
GSE135779 (n=44)	0.937 [0.910-0.958]	0.966 [0.950-0.982]	0.954 [0.925-0.974]
GSE174188 CD4 (n=261)	0.974 [0.969-0.976]	0.988 [0.985-0.990]	0.978 [0.968-0.982]
GSE285773 CD4 (n=26)	0.925 [0.840-0.975]	0.936 [0.872-0.978]	0.890 [0.779-0.950]

Table 2. Bidirectional source-only CD4-positive cohort transfer. Feature selection, scaling, PCA, and regularization were fitted in the source cohort and applied unchanged to the target cohort. Intervals are 5,000-resample donor-stratified bootstrap percentile intervals. Paired DeLong comparisons use the same target donors and are adjusted across four tests.

Source to target	Representation	ROC-AUC [95% CI]	PR-AUC	Brier score	Paired DeLong BH p vs Geneformer
GSE174188 CD4 to GSE285773 CD4	Frozen Geneformer	0.900 [0.762-0.994]	0.945	0.615	-
GSE174188 CD4 to GSE285773 CD4	Source-HVG pseudobulk	0.944 [0.844-1.000]	0.966	0.544	0.4914
GSE174188 CD4 to GSE285773 CD4	Source-PCA pseudobulk	0.981 [0.925-1.000]	0.990	0.180	0.1061
GSE285773 CD4 to GSE174188 CD4	Frozen Geneformer	0.884 [0.843-0.920]	0.936	0.237	-
GSE285773 CD4 to GSE174188 CD4	Source-HVG pseudobulk	0.919 [0.884-0.946]	0.954	0.155	0.0151
GSE285773 CD4 to GSE174188 CD4	Source-PCA pseudobulk	0.926 [0.895-0.953]	0.960	0.159	0.000603

Table 3. Leading methods in the recovered GSE174188 aggregation atlas. All methods used the same 261 donors, matched 500-cell estimand, and fixed five-fold donor split. Values are descriptive fixed-split metrics. The full 22-method ranking and all paired bootstrap intervals appear in Supplementary Figure S24 and the accompanying result table.

Method	Family	ROC-AUC	PR-AUC	Brier score
Raw pseudobulk	Expression	0.988	0.993	0.054
Expression PCA	Expression	0.986	0.992	0.054
Mean-HVG pseudobulk	Expression	0.984	0.991	0.055
scGPT mean pooling	Frozen embedding	0.978	0.988	0.062
Multiscale KME	Distribution summary	0.978	0.986	0.056
Attention MIL	Learned pooling	0.970	0.984	0.064
Weighted calibrated KME	Distribution summary	0.970	0.979	0.065
Count-calibrated KME	Distribution summary	0.970	0.978	0.064
Top-k MIL	Learned pooling	0.969	0.982	0.066
Mean plus variance	Distribution summary	0.967	0.965	0.066

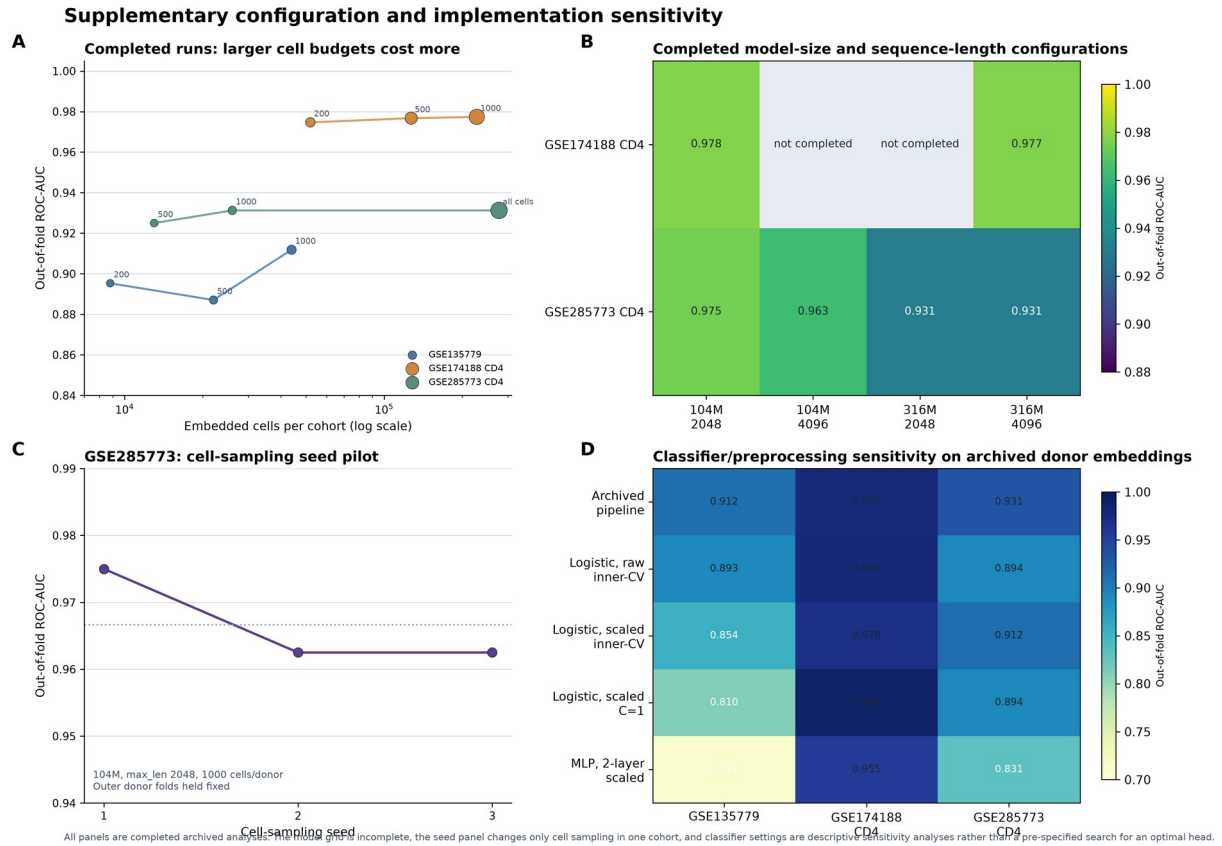
Table 4. Archived GSE174188 covariate sensitivity. Available covariates were cells per donor, mean UMI per cell, mean detected genes per cell, age, and sex. Residualization and downstream preprocessing were fitted within each training fold. Unadjusted values are from the matched archived fixed-split atlas.

Representation	Unadjusted ROC-AUC	Covariate-residualized ROC-AUC	Change
scGPT mean pooling	0.978	0.913	-0.065
Expression PCA	0.986	0.885	-0.100
Mean-HVG pseudobulk	0.984	0.916	-0.068
Technical/demographic covariates only	-	0.789	-

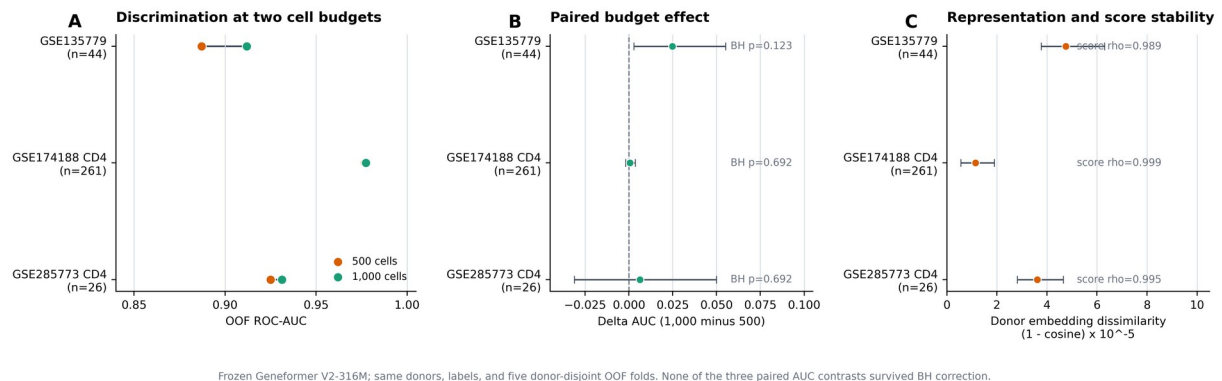
Table 5. Fixed Geneformer pooling under reciprocal source-only transfer at 500 cells per donor. Parentheses give the AUC difference from coordinate mean in the same target. Paired DeLong q values are Benjamini-Hochberg adjusted across seven candidates and two directions (14 tests). The continuation rule required an AUC gain of at least 0.010 in the 261-donor target, no loss in the 26-donor target, and the prespecified calibration bounds.

Pooling operator	Target n=26 AUC (difference)	q	Target n=261 AUC (difference)	q	Continued to cap1000
Coordinate mean	0.863 (reference)	-	0.884 (reference)	-	-
Coordinate median	0.919 (+0.056)	0.306	0.858 (-0.026)	5.59e-4	No
10% trimmed mean	0.856 (-0.006)	0.745	0.892 (+0.008)	0.0960	No
Mean plus standard deviation	0.825 (-0.037)	0.661	0.885 (+0.001)	0.923	No
Source-PCA32 quantiles	0.781 (-0.081)	0.371	0.823 (-0.062)	0.00604	No
Source-PCA16 moments	0.925 (+0.063)	0.307	0.839 (-0.046)	0.0613	No
Source-PCA8 donor mean	0.944 (+0.081)	0.307	0.834 (-0.050)	8.38e-4	No
Source-PCA16 donor mean	0.913 (+0.050)	0.478	0.784 (-0.100)	8.79e-6	No

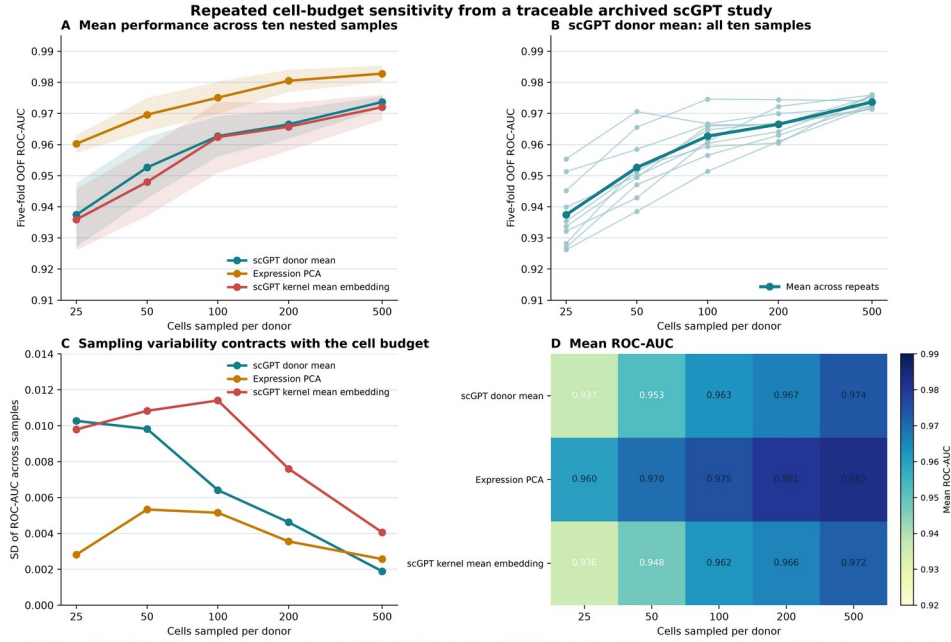
Supplementary figures



Supplementary Figure S1. Configuration and implementation sensitivity. Panel A relates max_len4096 embedded-cell counts to observed AUC and wall-clock GPU time (point area). Panel B displays the available portions of the model-size and maximum-sequence-length grid. Panel C varies cell-sampling seed while holding outer donor folds fixed in GSE285773. Panel D shows classifier and preprocessing sensitivity on donor embeddings.

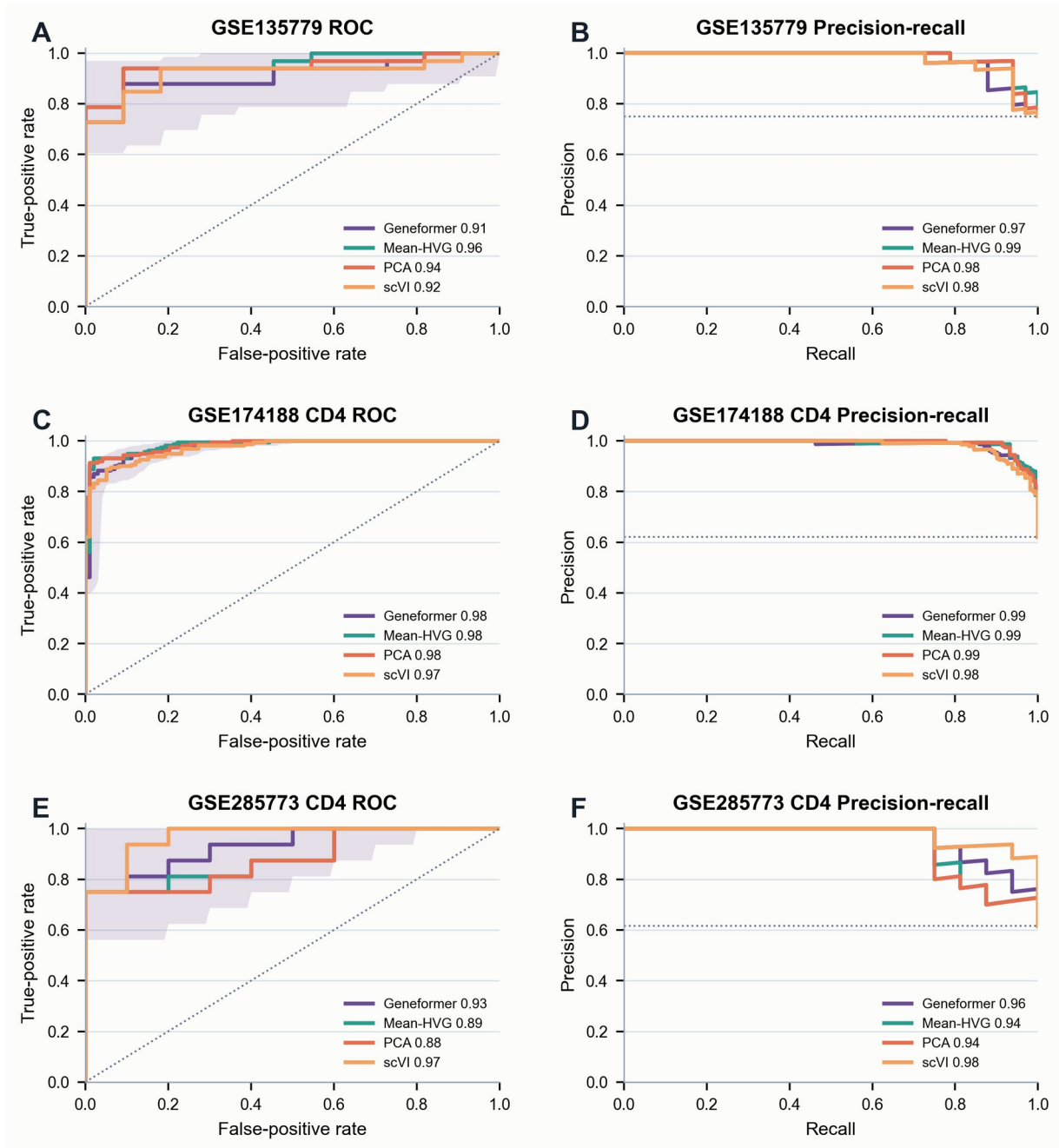


Supplementary Figure S2. Matched Geneformer cell-budget comparison. The analysis uses the same donors, labels, and folds at 500 and 1000 cells per donor. Panel A shows OOF AUC; panel B shows the paired 1000-minus-500 AUC difference with bootstrap intervals and BH-adjusted paired DeLong p values; panel C reports donor-embedding cosine dissimilarity and donor-score rank correlation.

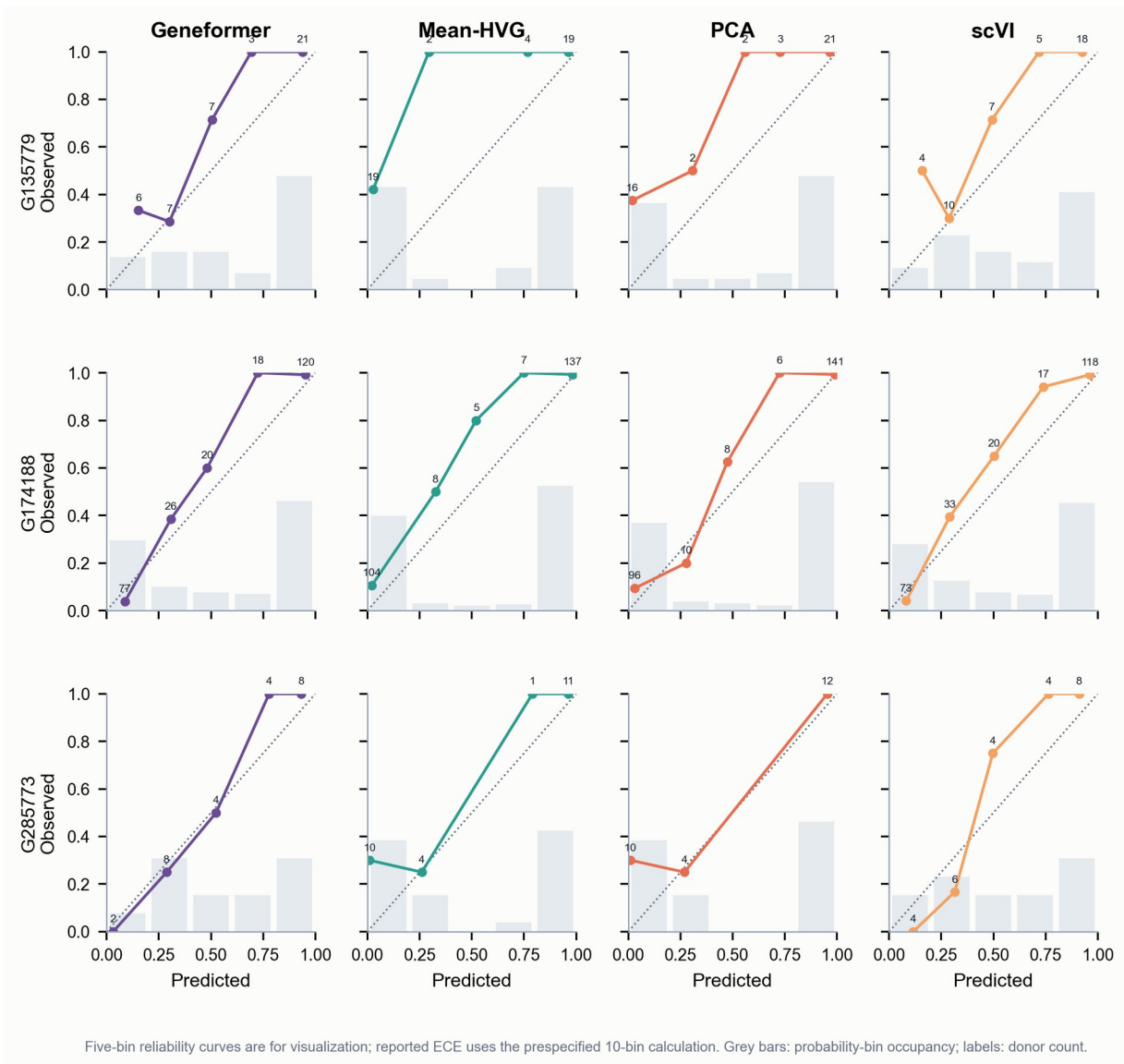


Fixed common-support donor set: 237 donors (141 SLE, 96 control); nested samples; ten repeats per budget; five-fold OOF evaluation. scGPT 0.2.4 checkpoint, vocabulary, configuration, embedding, and fold records are retained; not a direct Geneformer comparison.

Supplementary Figure S3. scGPT low-cell-count sensitivity in GSE174188 CD4-positive cells. A 237-donor set (141 SLE and 96 controls) was evaluated with five donor-level folds and ten nested within-donor samples at each cap. Panels A-C show mean performance, individual scGPT donor-mean trajectories, and between-sample variability; panel D shows mean AUC across representations. The analysis used scGPT 0.2.4 with the whole-human checkpoint and provides an independent low-cell-count analysis.



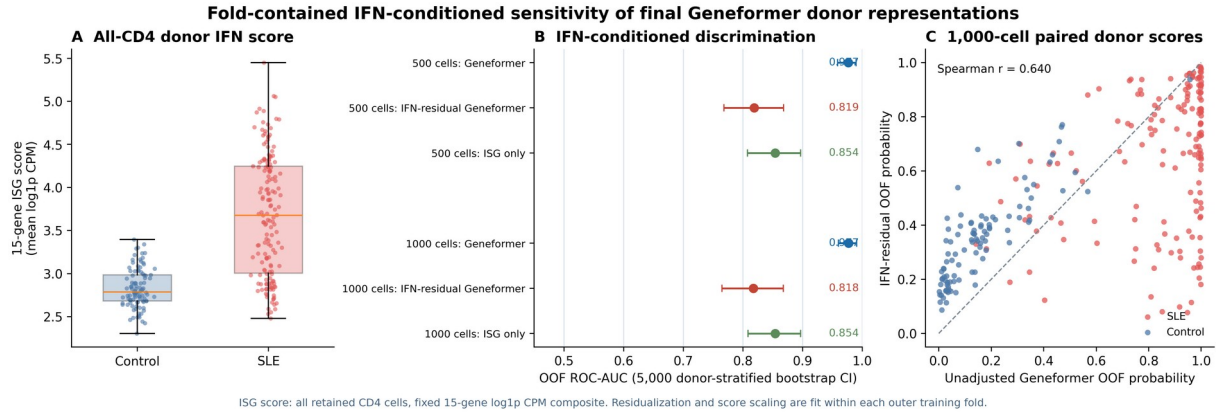
Supplementary Figure S4. Fixed-split ROC and precision-recall curves. Curves use donor-level out-of-fold scores for Geneformer, mean-HVG pseudobulk, PCA pseudobulk, and scVI donor means in each cohort. Shaded Geneformer ROC ribbons are 300-resample stratified donor bootstrap visualizations. The repeated internal comparison is shown in Figure 2 and Table 1.



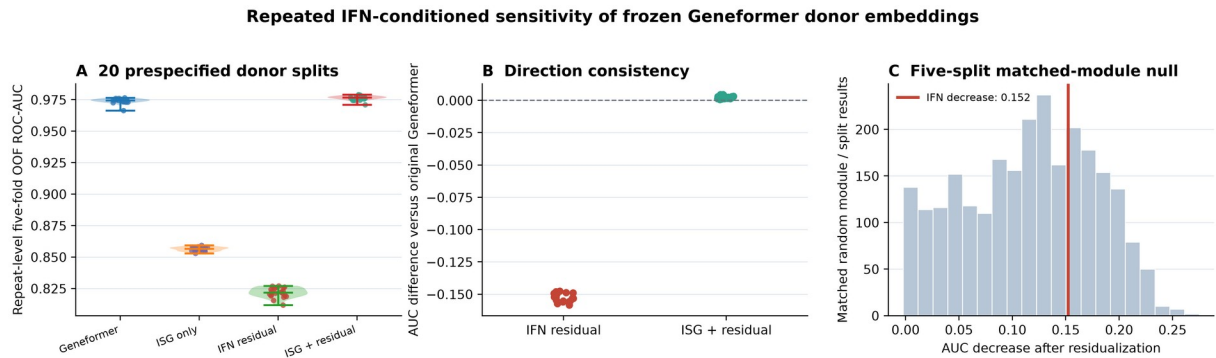
Supplementary Figure S5. Within-cohort calibration. Reliability curves use five display bins, and the ECE calculation uses 10 equal-width bins. Grey bars show bin occupancy and labels give donor counts.



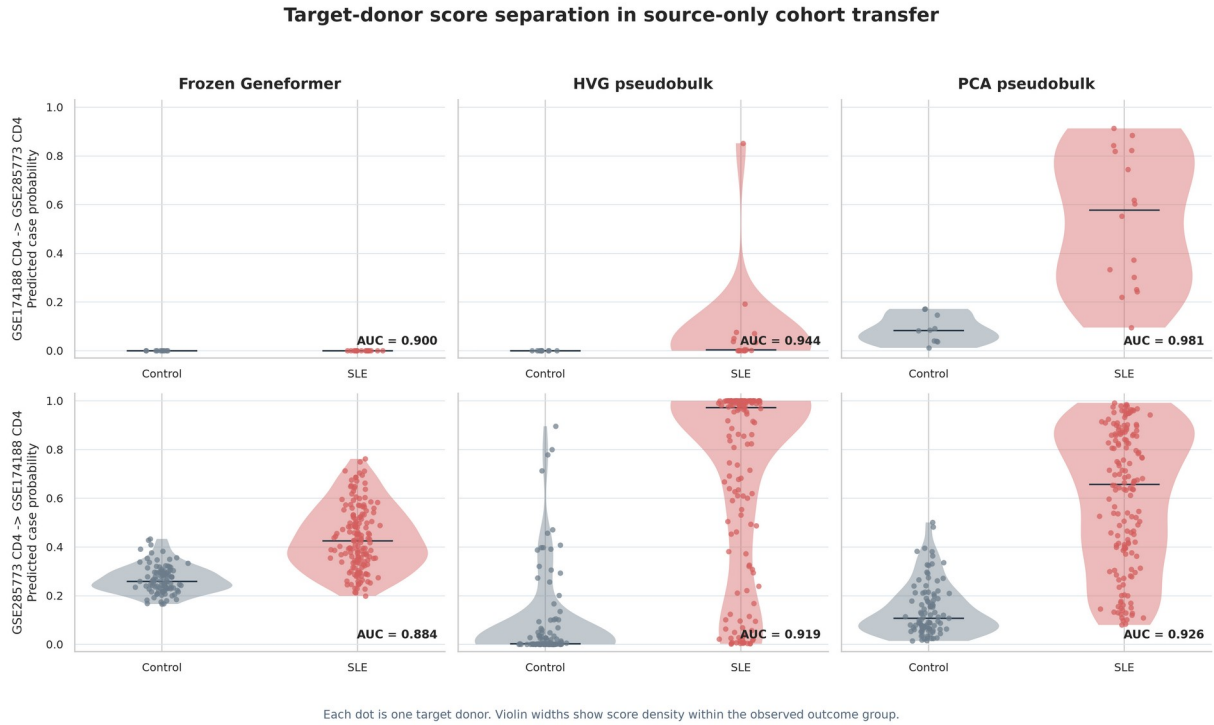
Supplementary Figure S6. Cohort-specific pseudobulk expression rankings. Genes are ordered by absolute SLE-minus-control donor-pseudobulk log1p CPM difference. Paired panels show two-sided donor-level Welch test evidence, capped at $-\log_{10} P = 50$.



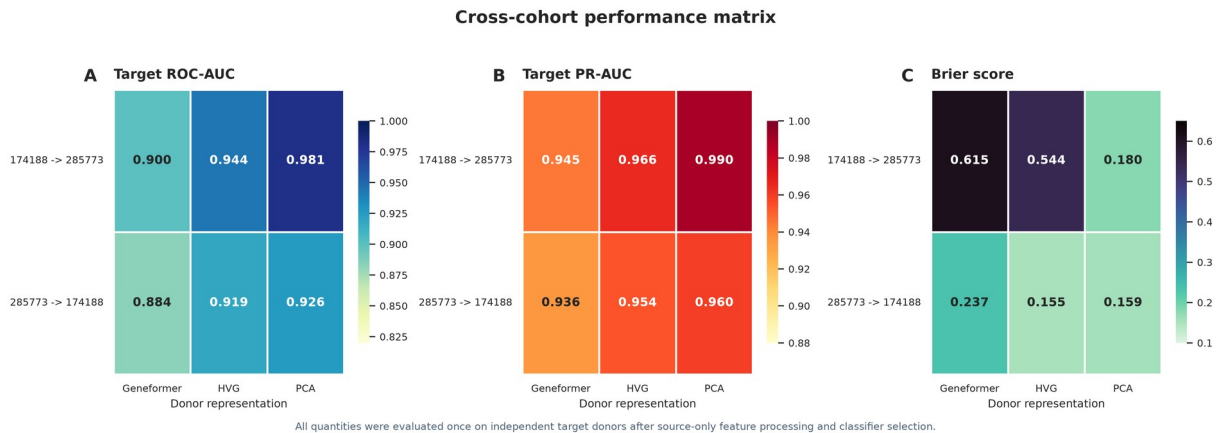
Supplementary Figure S7. Interferon-conditioned Geneformer sensitivity in GSE174188 CD4-positive cells. Panel A shows the fixed 15-gene all-CD4 donor ISG score. Panel B compares Geneformer, ISG-only, and ISG-residualized OOF AUC at 500 and 1000 cells. Panel C compares paired unadjusted and residualized 1000-cell donor probabilities. ISG score scaling and the coordinate-wise residual models were fitted within each outer training fold. ISG scores used all available CD4 cells per donor. Supplementary Figure S8 shows repeated-split and matched-module results.



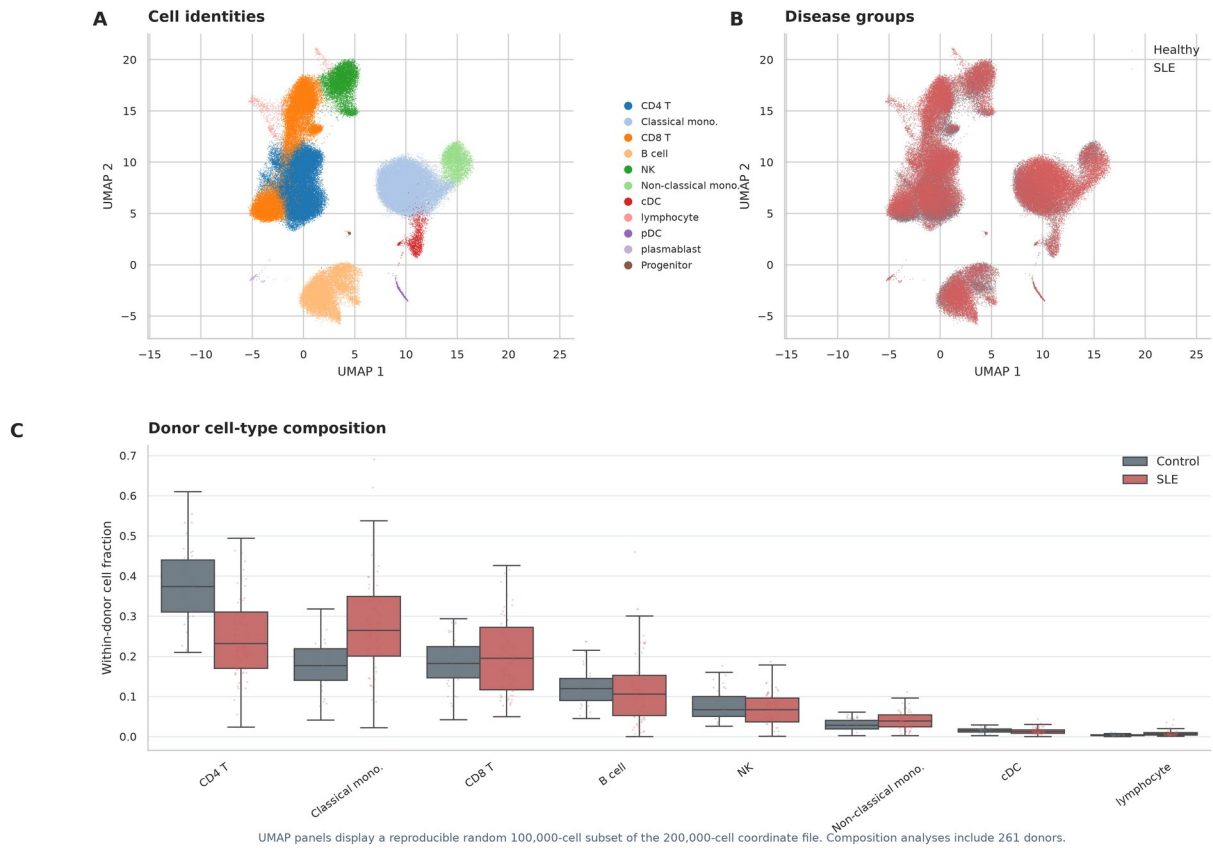
Supplementary Figure S8. Repeated IFN-conditioned sensitivity and matched-module comparison. Panel A shows 20 repeated donor-stratified five-fold splits for Geneformer, ISG-only, IFN-residual, and ISG-plus-residual models. Panel B shows the change from Geneformer in every split. Panel C compares the median IFN-associated decrease with 500 expression-, variance-, score-variance-, and coexpression-matched non-ISG modules. The stable IFN association was comparable to that of other structured expression modules.



Supplementary Figure S9. Target-donor score distributions in source-only cohort transfer. Each dot is one target donor, with color indicating the observed outcome. The six panels show the distribution of predicted case probabilities for every transfer direction and donor representation; values are the corresponding target-cohort ROC-AUCs.

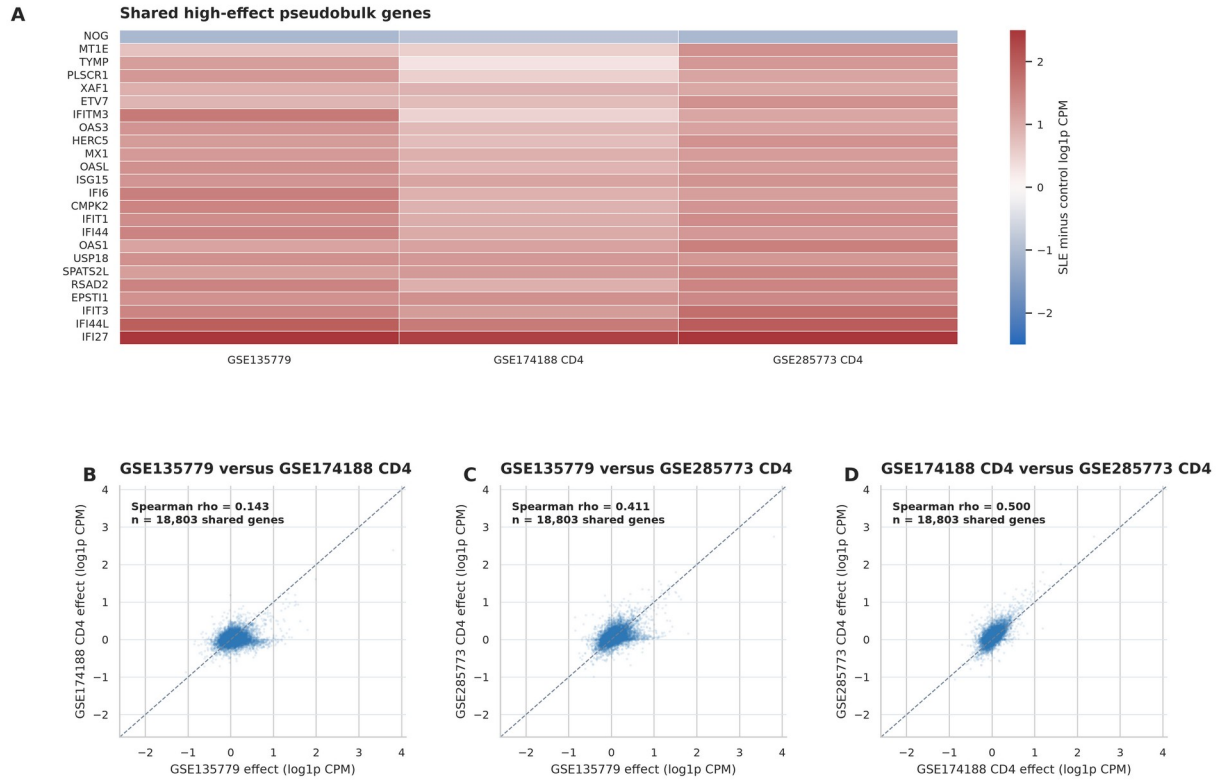


Supplementary Figure S10. Cross-cohort performance matrix. ROC-AUC, PR-AUC, and Brier score are shown for the six source-only transfer evaluations. Each quantity was calculated on the independent target cohort after source-cohort feature processing and classifier selection.

GSE174188 cellular landscape and donor composition

Supplementary Figure S11. GSE174188 cellular landscape and donor composition. Panels A-B show a reproducible random 100,000-cell subset of the 200,000-cell UMAP coordinate resource colored by cell identity and disease group. Panel C descriptively summarizes the eight most abundant cell types across 261 donors. This whole-PBMC landscape is not used as a matched CD4-only classifier baseline.

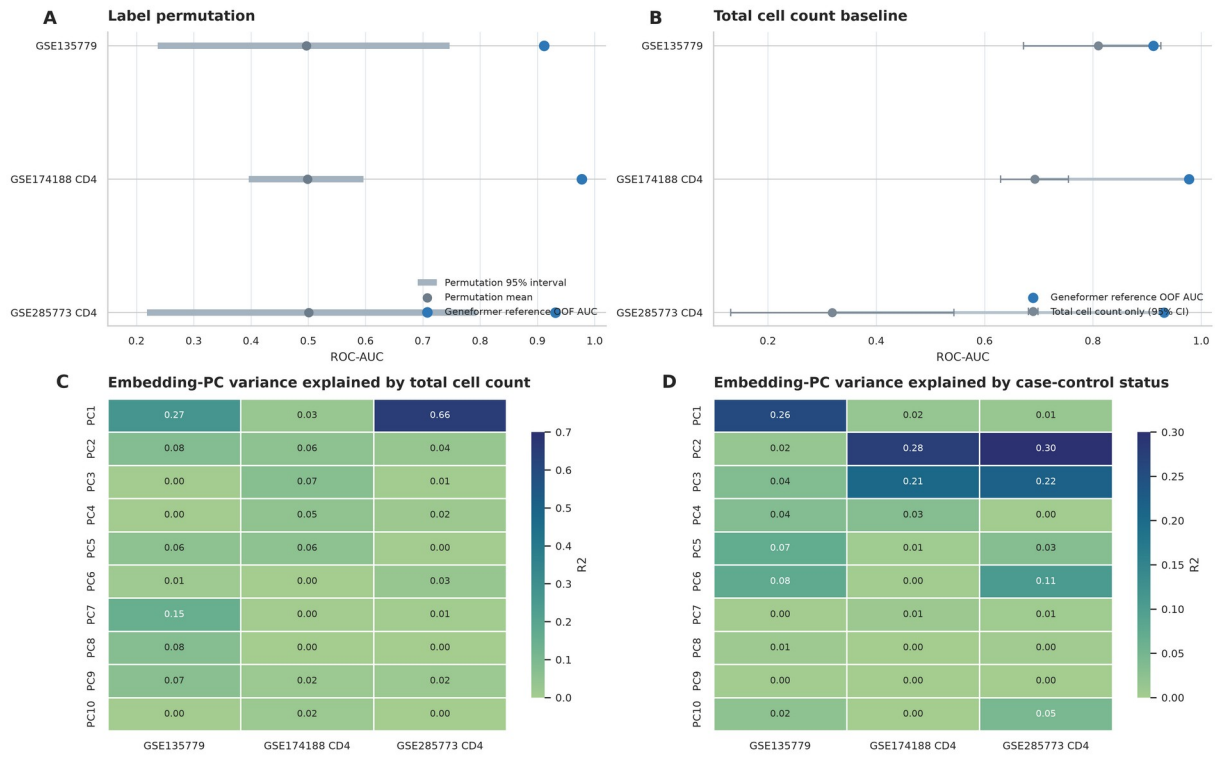
Recurring pseudobulk SLE expression effects across cohorts



Panel A shows the 24 shared genes with the largest mean absolute cohort effect. Panels B-D show all genes shared by each cohort pair.

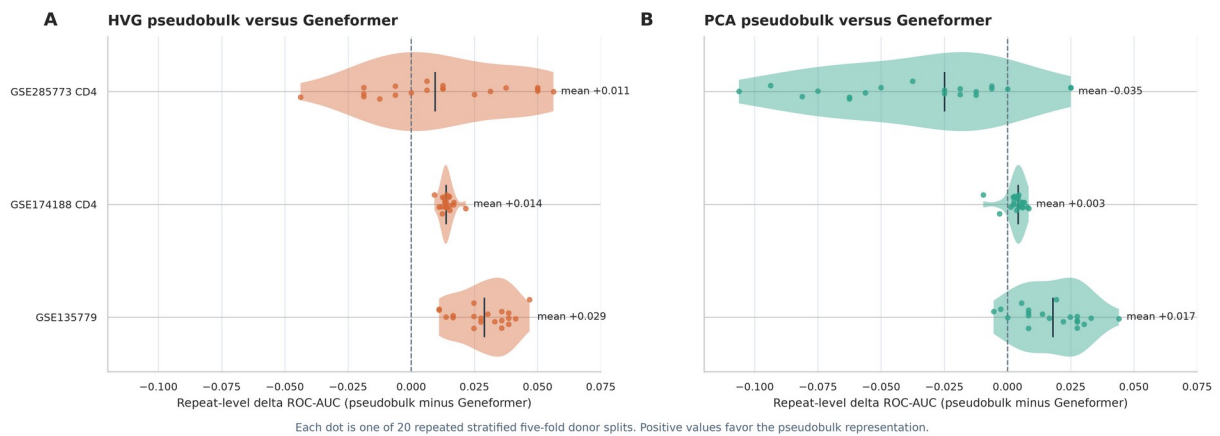
Supplementary Figure S12. Recurring pseudobulk SLE expression effects across cohorts. Panel A displays the 24 genes with the largest mean absolute SLE-minus-control log1p CPM difference among genes shared by all three cohorts. Panels B-D display full shared-gene effect-size comparisons for each cohort pair.

Donor-level Geneformer signal and technical context



Supplementary Figure S13. Donor-level Geneformer signal and technical context. Panel A contrasts fixed-split Geneformer AUCs with label-permutation distributions. Panel B displays total-cell-count and Geneformer AUCs. Panels C-D show univariate R² values for the first ten donor-embedding principal components explained by total cell count and case-control status.

Paired repeated donor-level performance differences

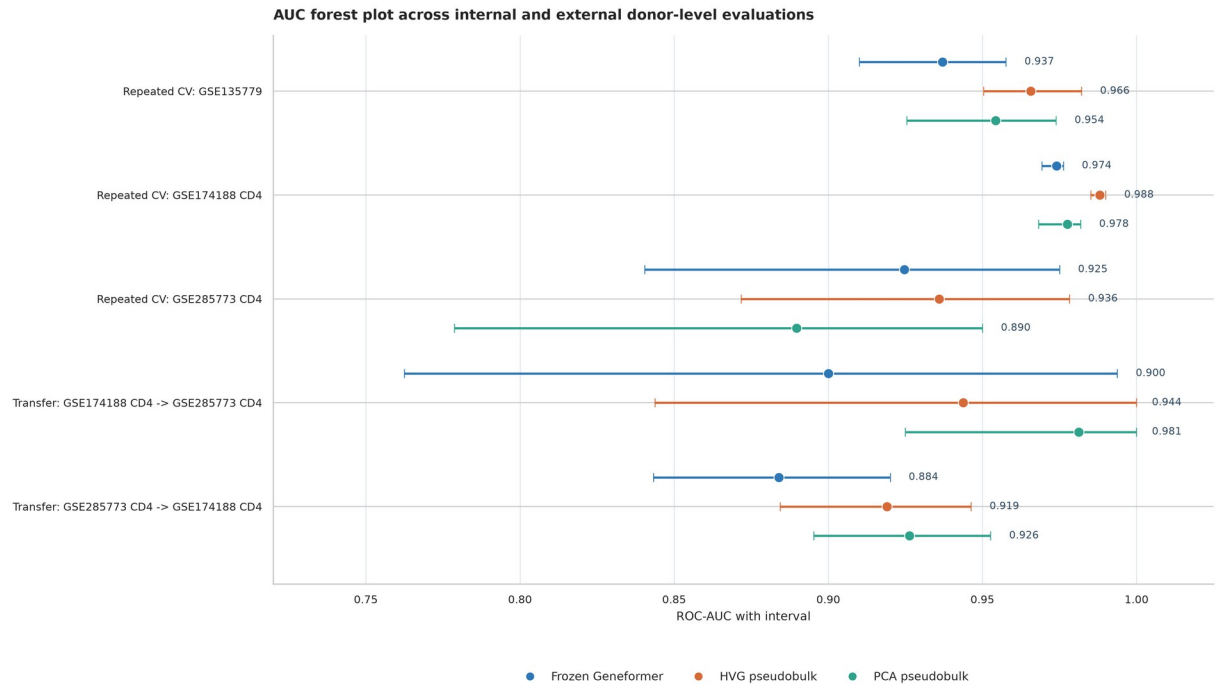


Supplementary Figure S14. Paired repeated donor-level performance differences. Each point is one of 20 repeated stratified five-fold donor splits. Positive values indicate a higher ROC-AUC for the pseudobulk representation than for Geneformer on the same donor split.



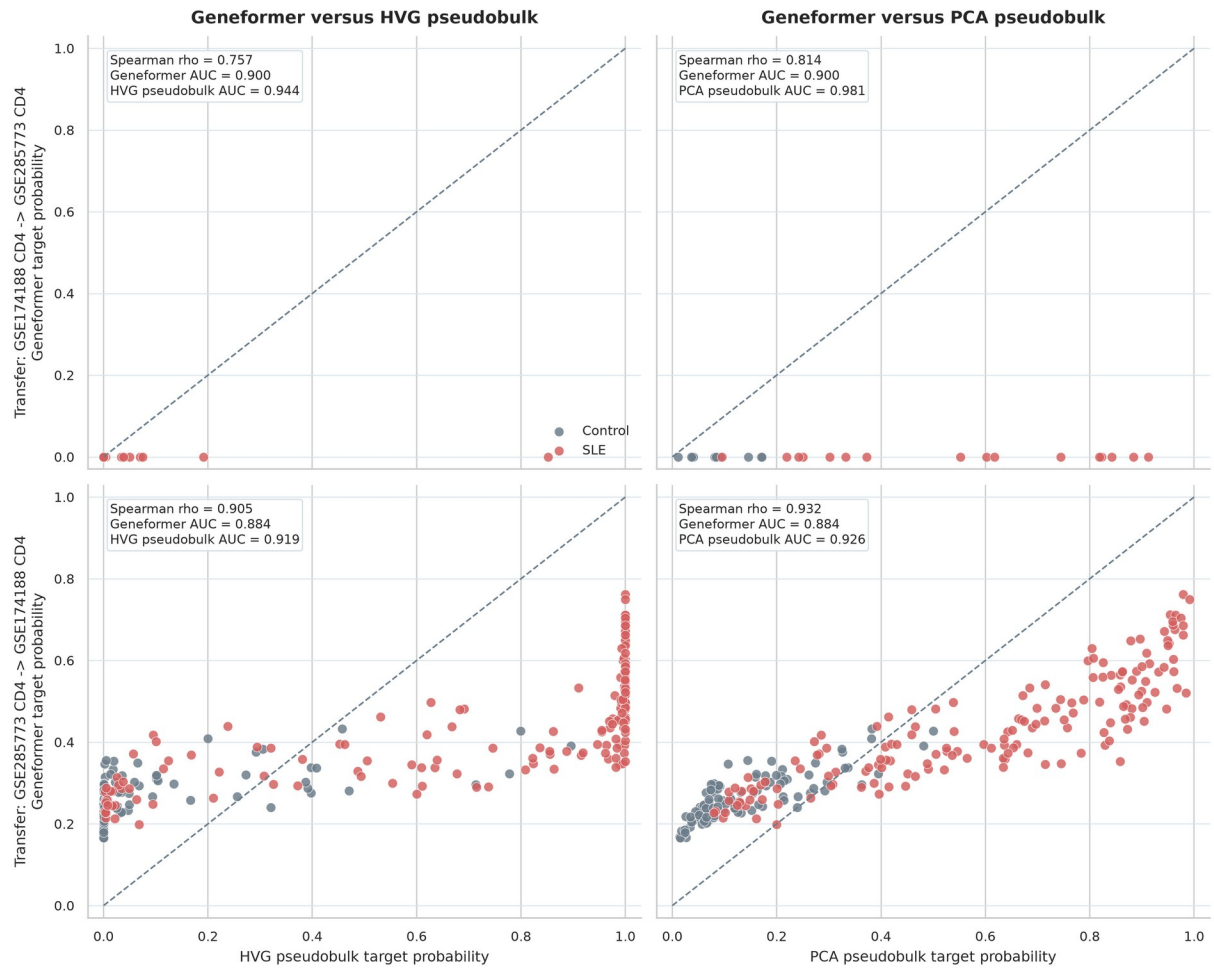
Each representation was projected independently with unsupervised UMAP; case-control labels were used only for colour. HVG and PCA pseudobulk share a donor-expression input but retain different feature geometries.

Supplementary Figure S15. Donor-level representation geometry. Each donor is projected from its full Geneformer, HVG-pseudobulk, or PCA-pseudobulk representation in an independently fitted, label-blind UMAP. Case-control labels are used only for color. The panel is descriptive and visualizes each representation's donor geometry rather than a supervised decision boundary.



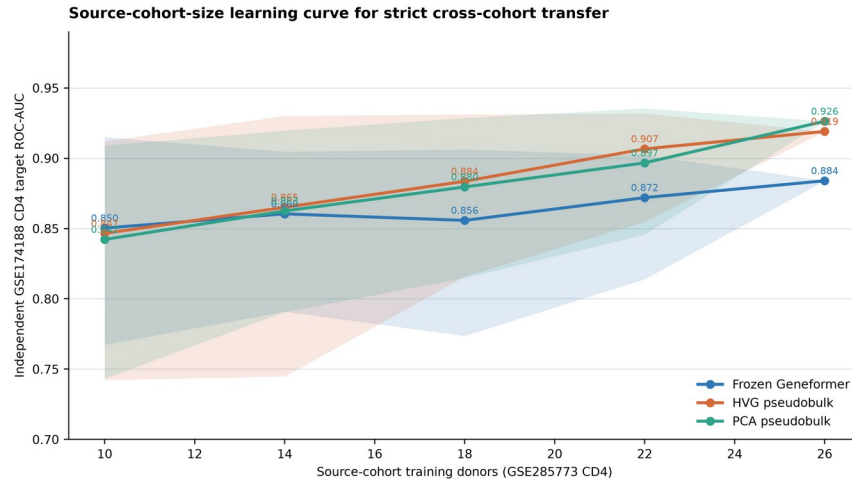
Internal intervals are the 2.5th-97.5th percentiles across 20 repeated stratified donor folds; transfer intervals are 5,000-resample stratified bootstrap 95% CIs on independent target donors.

Supplementary Figure S16. AUC forest plot across donor-level evaluations. The panel summarizes all three representations in the three repeated within-cohort analyses and two strict source-only transfer directions. Internal intervals are the 2.5th-97.5th percentiles over 20 repeated stratified donor-fold analyses; transfer intervals are 5,000-resample donor-stratified bootstrap 95% confidence intervals on independent target donors.

Paired target-donor prediction agreement under source-only transfer

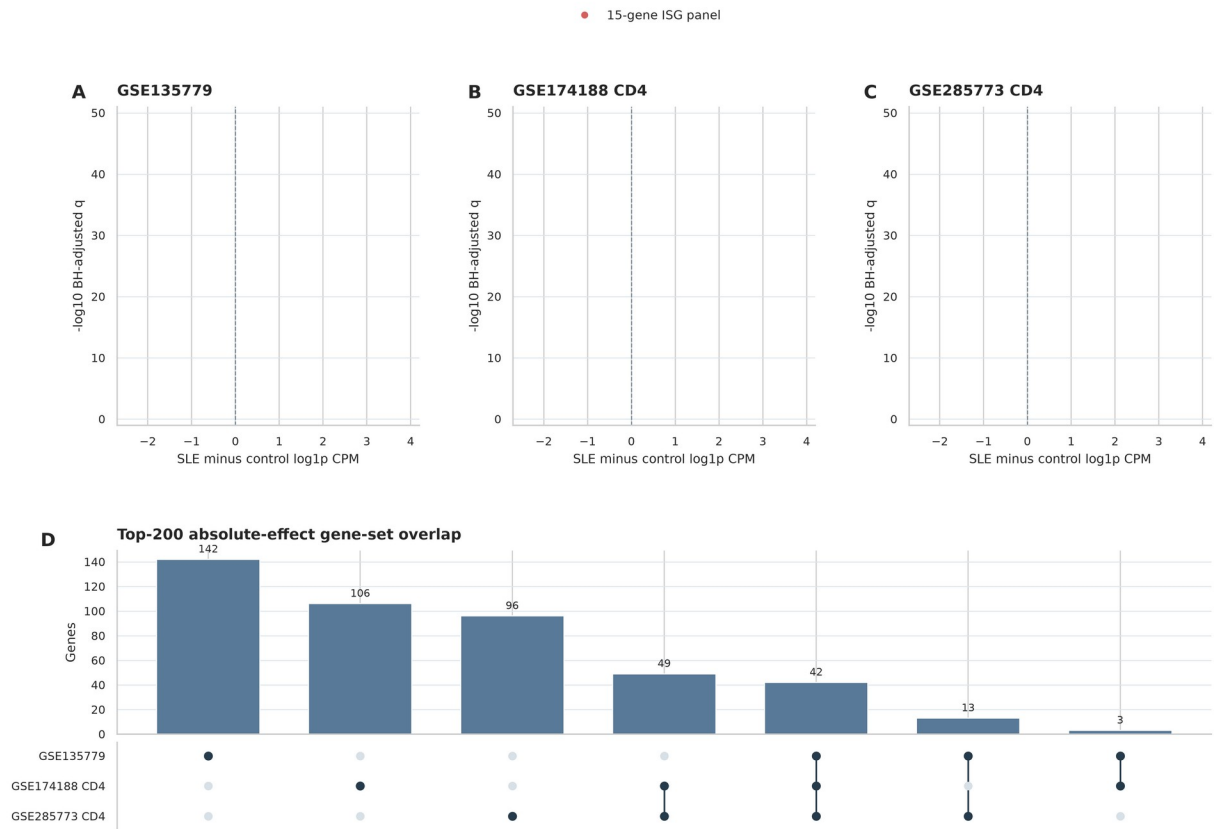
Every point is the same independent target donor scored by both representations. The diagonal indicates equal predicted case probabilities.

Supplementary Figure S17. Paired target-donor score agreement. Each point is one independent target donor scored by Geneformer and by the indicated pseudobulk representation under the same source-only transfer model. The diagonal indicates equal predicted probabilities; annotations give target-donor rank correlation and ROC-AUCs.

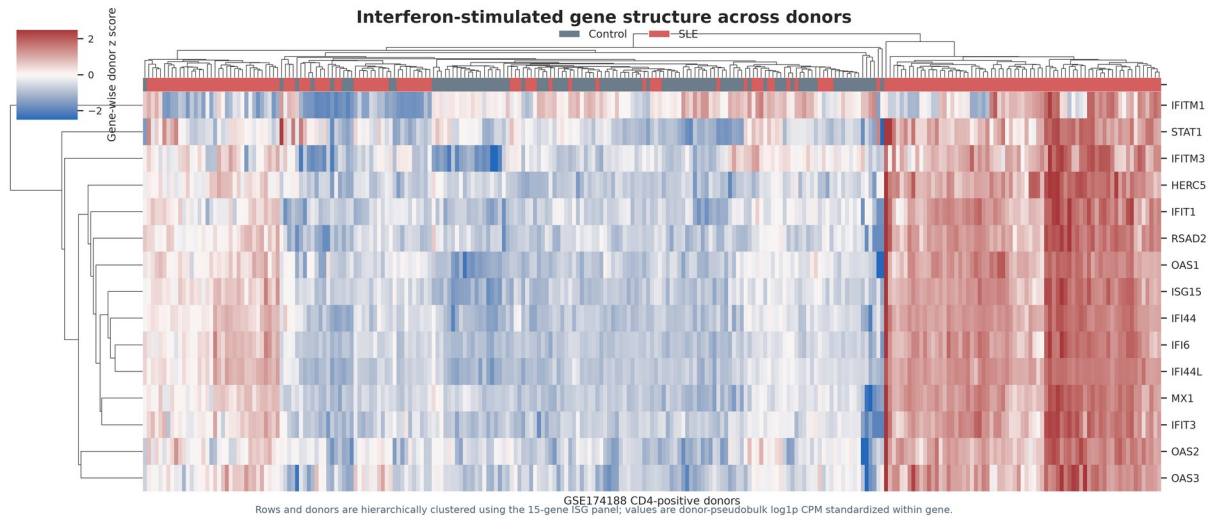


For $n < 26$, points are means across 40 stratified source-donor subsets and bands are 2.5th-97.5th percentiles. The $n = 26$ endpoint is the single prespecified full-source analysis and therefore has no resampling band.

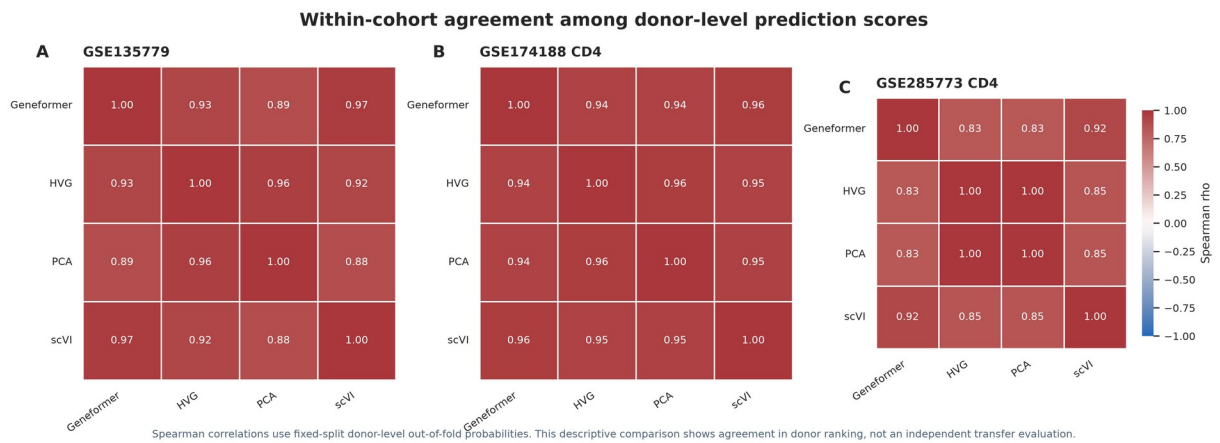
Supplementary Figure S18. Source-cohort-size learning curve for strict transfer. GSE285773 CD4-positive donors were repeatedly subsampled at $n=10$, 14 , 18 , and 22 , with feature selection, scaling, PCA, and logistic-regularization selection repeated within each source subset. Curves report mean independent-target ROC-AUC across 40 subsets; bands are 2.5th-97.5th percentiles on the fixed 261-donor GSE174188 target. The $n=26$ endpoint is the single prespecified full-source analysis and has no resampling band.

Pseudobulk SLE effects and cross-cohort gene overlap

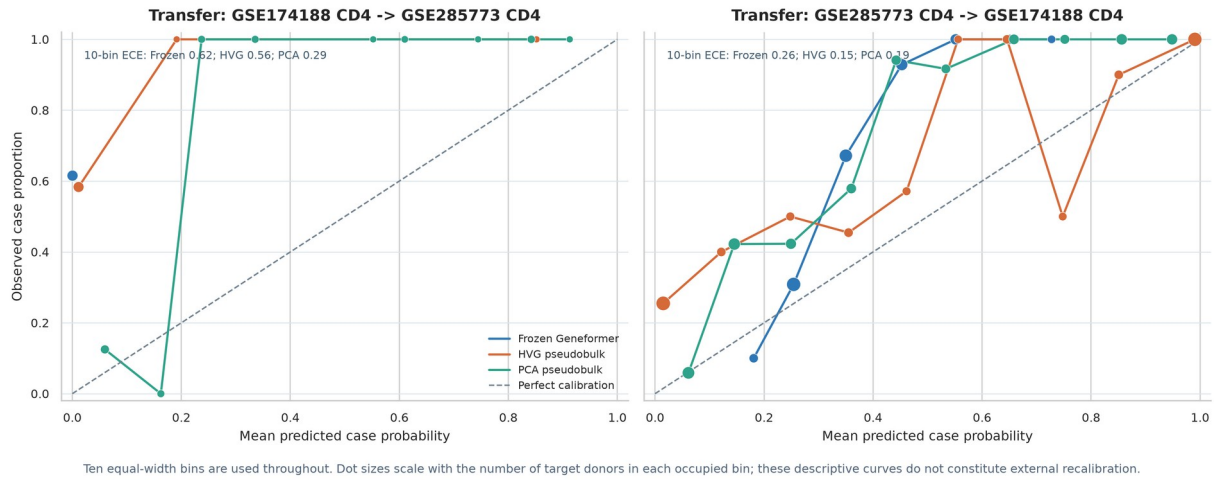
Supplementary Figure S19. Pseudobulk volcano plots and high-effect gene overlap. Panels A-C display donor-pseudobulk SLE-control effect sizes and Benjamini-Hochberg-adjusted two-sided Welch-test evidence. The red points are the fixed 15-gene ISG panel. Panel D is an UpSet-style display of membership in the top 200 absolute-effect genes from each cohort.



Supplementary Figure S20. ISG expression across GSE174188 donors. Rows and donors are hierarchically clustered using the fixed 15-gene ISG panel. Values are donor-pseudobulk log1p CPM standardized within each gene; the top annotation records the observed case-control group.

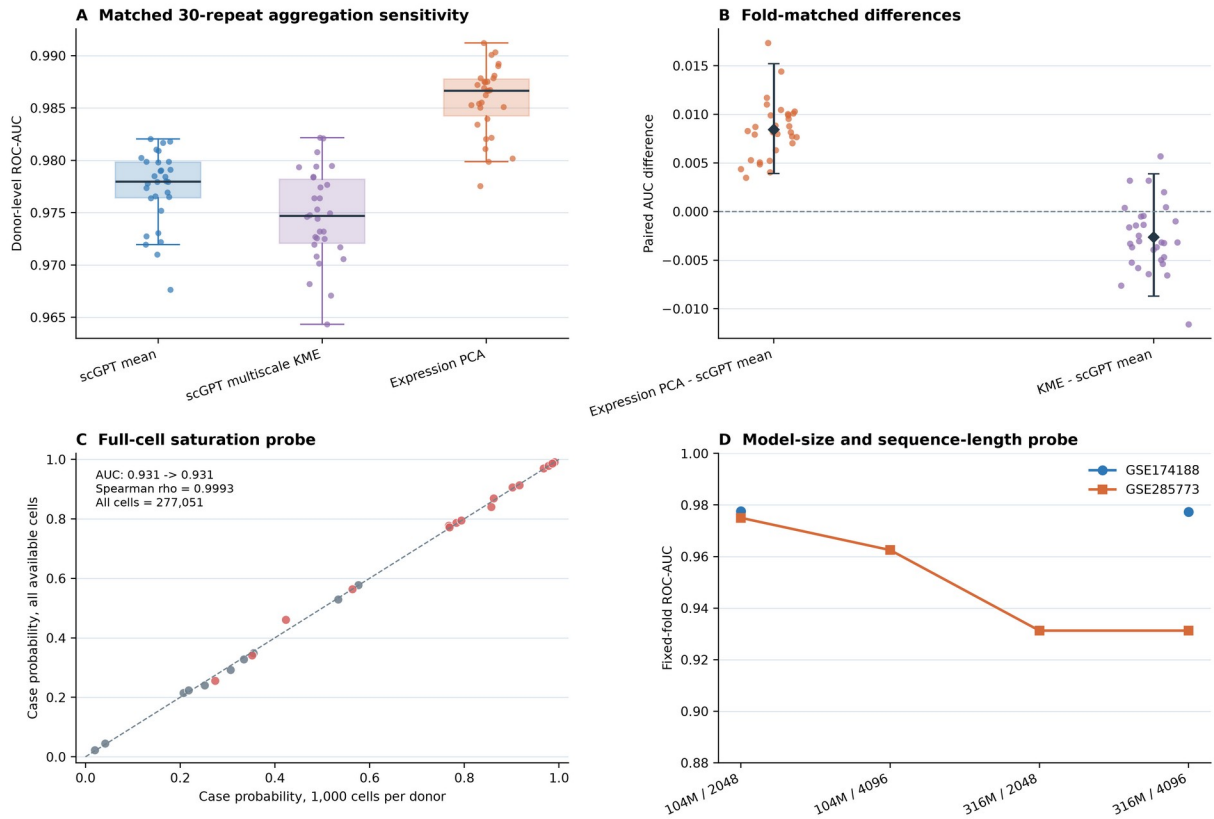


Supplementary Figure S21. Within-cohort agreement among donor prediction scores. Spearman correlations are calculated from fixed-split donor-level out-of-fold probabilities for Geneformer, HVG pseudobulk, PCA pseudobulk, and scVI donor means. This is a descriptive agreement analysis rather than an independent transfer comparison.

Target-cohort reliability under source-only transfer

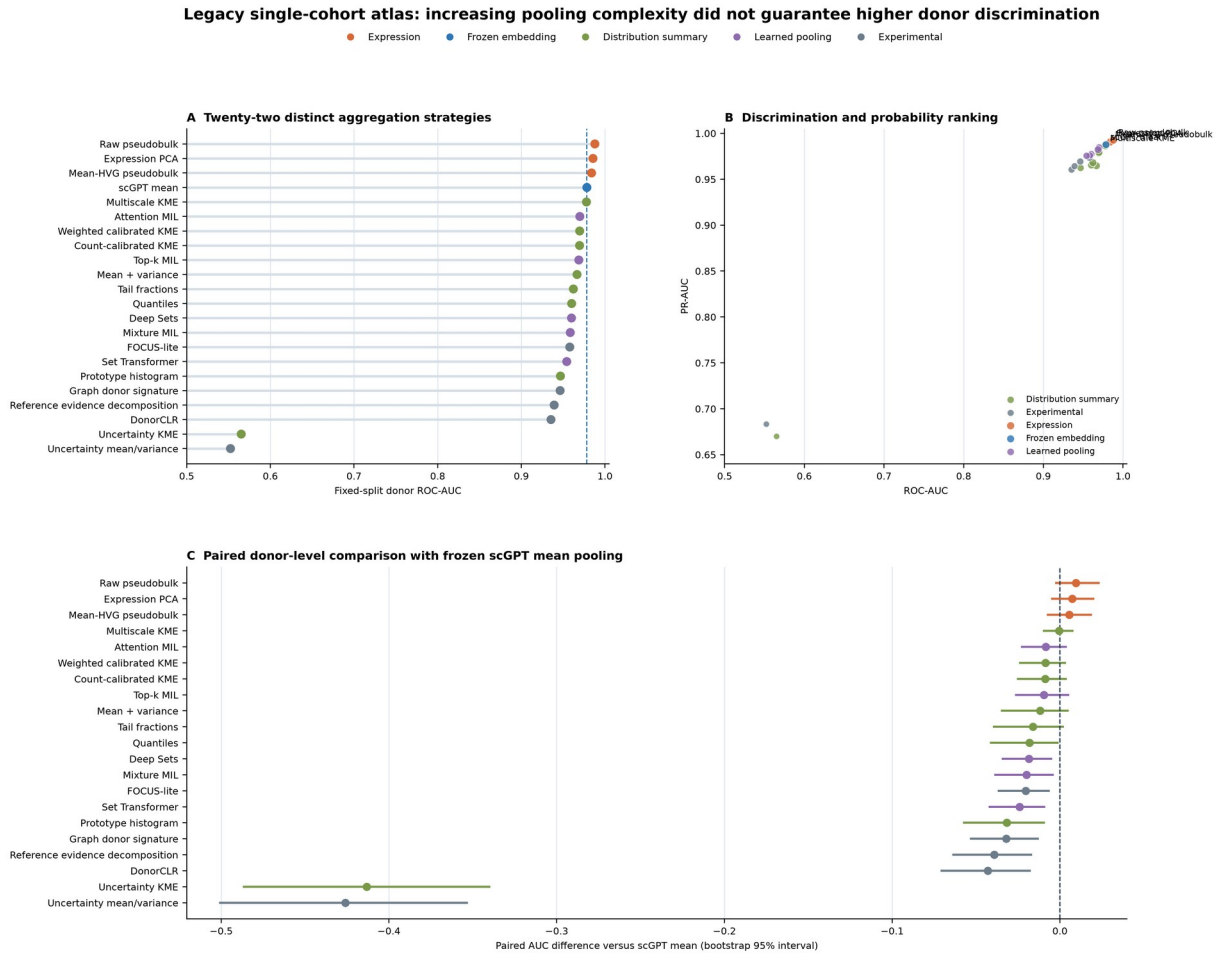
Supplementary Figure S22. Target-cohort reliability in strict transfer. Reliability curves use the same 10 equal-width bins as the reported ECE values; dot size indicates the number of target donors in each occupied bin. The curves are descriptive target-cohort diagnostics and do not constitute external recalibration.

Archived sensitivity analyses bound aggregation and compute choices



Panels A-B: GSE174188 CD4, 261 donors, 500 cells per donor, matched donor-disjoint five-fold repeats. Panels C-D are fixed-fold sensitivity probes and are interpreted descriptively.

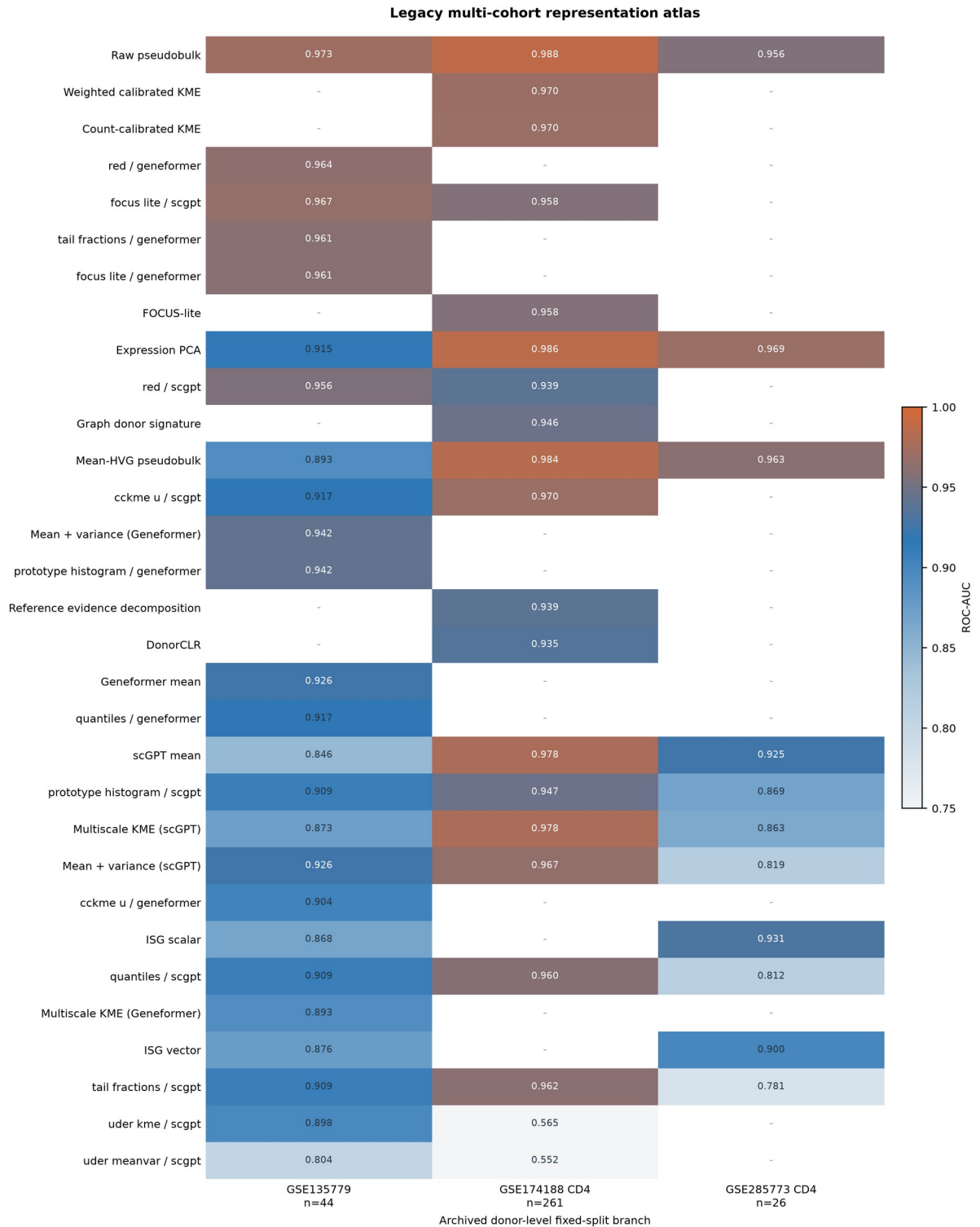
Supplementary Figure S23. Archived aggregation and scaling sensitivities. Panels A-B compare scGPT mean pooling, multiscale kernel mean embedding, and expression PCA across 30 matched donor-disjoint five-fold repeats in 261 GSE174188 donors. Panel C compares the 1,000-cell Geneformer run with all 277,051 available GSE285773 cells. Panel D reports the incomplete fixed-fold model-size and sequence-length grid. Repeat differences and configuration probes are interpreted descriptively.



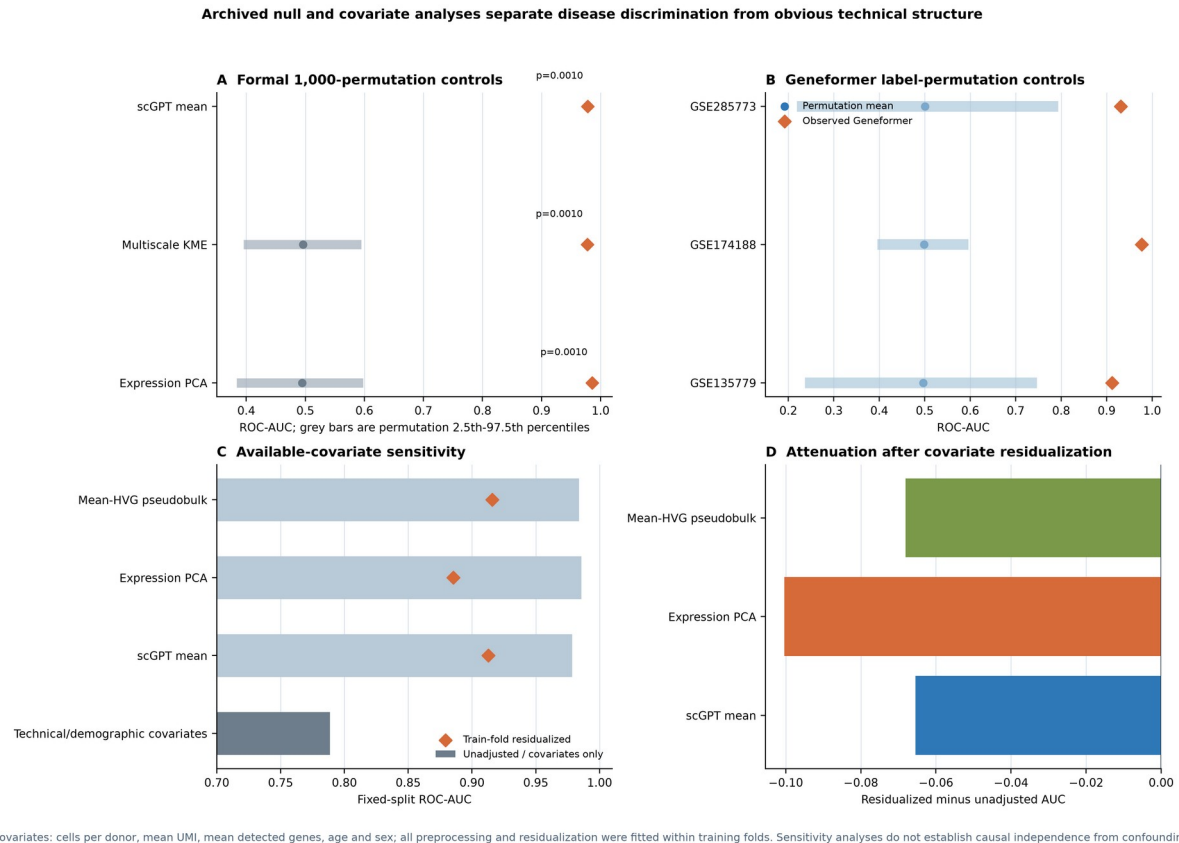
GSE174188 CD4, 261 donors, matched 500-cell estimand, one fixed five-fold donor split. This broad atlas is exploratory and is not an external-transfer comparison.

Supplementary Figure S24. Broad fixed-split donor-pooling atlas. Panel A ranks 22 distinct completed representations in the 261-donor GSE174188 matched 500-cell analysis. Panel B relates ROC-AUC and PR-AUC, with point area reflecting lower Brier score. Panel C shows paired donor-bootstrap AUC differences from scGPT mean pooling. Exact duplicate aliases were removed from display. The atlas uses one donor split and is exploratory rather than an external validation.

Cross-cohort SLE donor representations

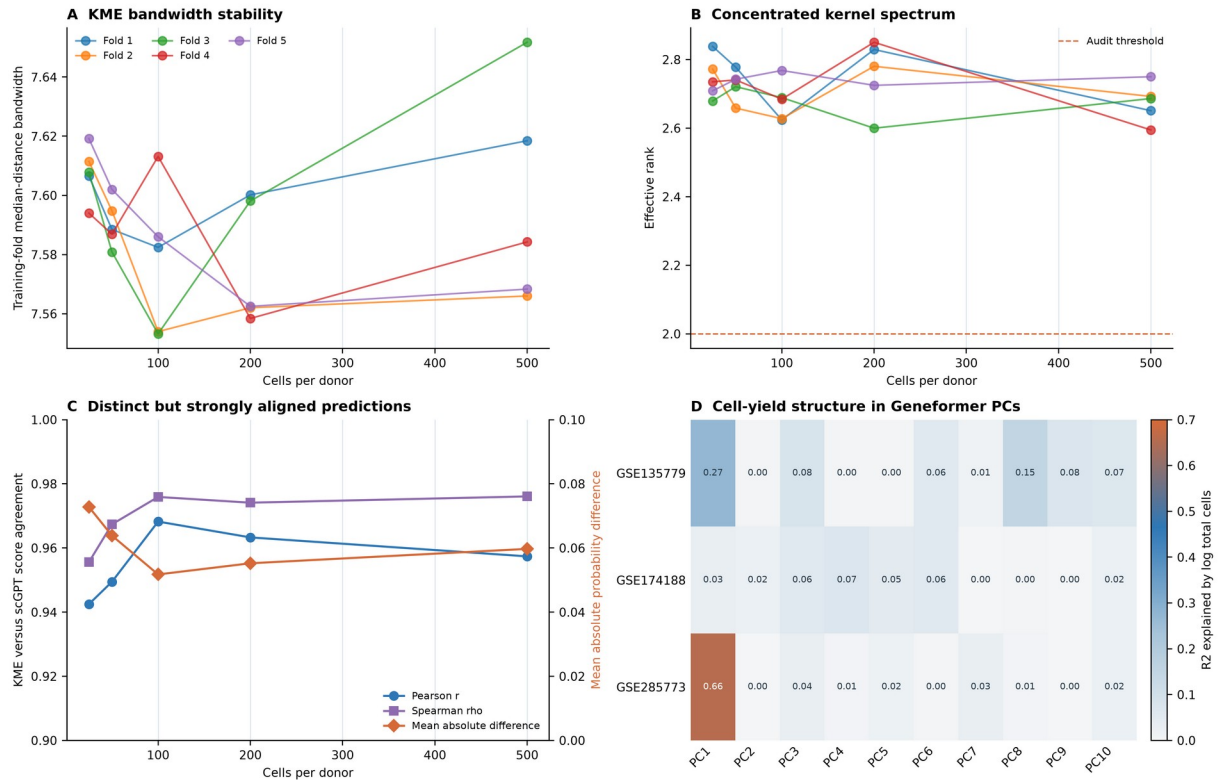


Supplementary Figure S25. Historical multi-cohort representation atlas. The heatmap assembles 58 traceable completed method records from GSE135779, GSE174188 CD4, and GSE285773 CD4. Blank cells denote configurations that were not run. Because fold type, model availability, and historical implementation branch differ across cells, this figure visualizes coverage and broad performance patterns but does not support formal average-rank inference.



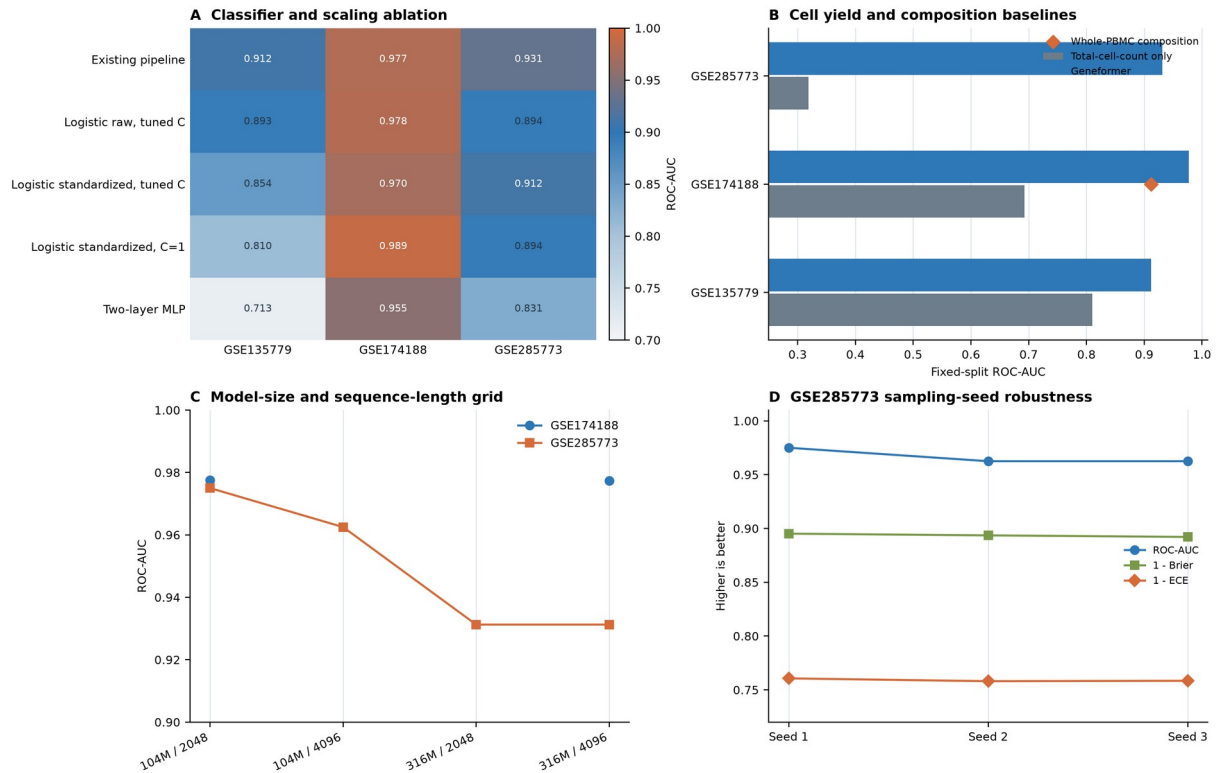
Supplementary Figure S26. Formal null and available-covariate sensitivity analyses. Panel A reports accepted 1,000-permutation tests for scGPT mean, expression PCA, and multiscale KME in GSE174188; the recovered scGPT record retains the accepted observed AUC and empirical p value, while its null quantiles were not present in the local snapshot. Panel B shows archived 1,000-permutation Geneformer controls in all three cohorts. Panels C-D compare fixed-split representation AUCs before and after training-fold residualization for cells per donor, mean UMI, mean detected genes, age, and sex.

Kernel and embedding diagnostics expose where stable representations remain structurally narrow



KME diagnostics use 237 common-support donors and one sampling repeat. PC associations are univariate descriptive R2 values and do not adjust for correlated technical or clinical factors.

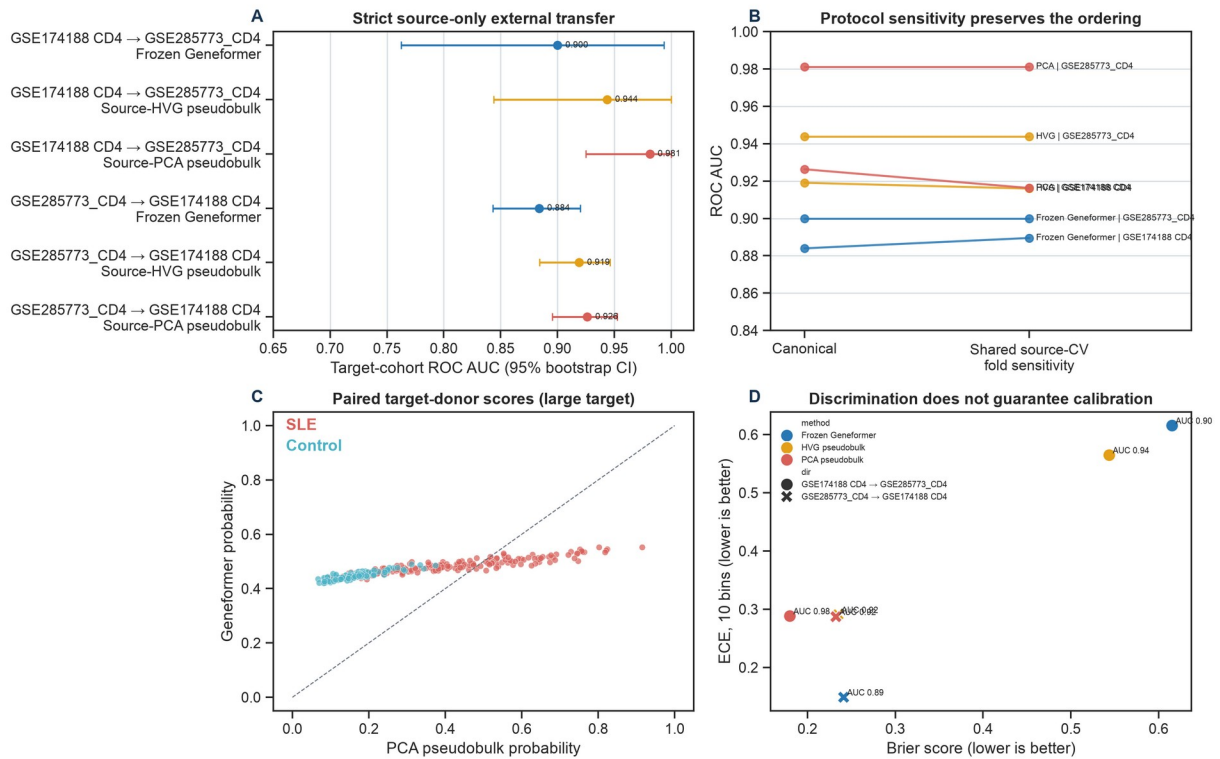
Supplementary Figure S27. Kernel and embedding diagnostics. Panels A-B show fold-level multiscale-KME bandwidth and effective rank across 25-500 cells per donor. Panel C compares KME and scGPT donor probabilities; no donor scores were exactly identical. Panel D shows univariate R2 values for the first ten Geneformer donor-embedding PCs explained by log total cell yield. The kernel spectrum was stable but strongly concentrated.

Engineering and nuisance-variable probes bound the fixed Geneformer pipeline

Composition uses the whole-PBMC GSE174188 resource and is therefore an auxiliary, unmatched baseline. Configuration and seed analyses are incomplete fixed-fold sensitivity grids.

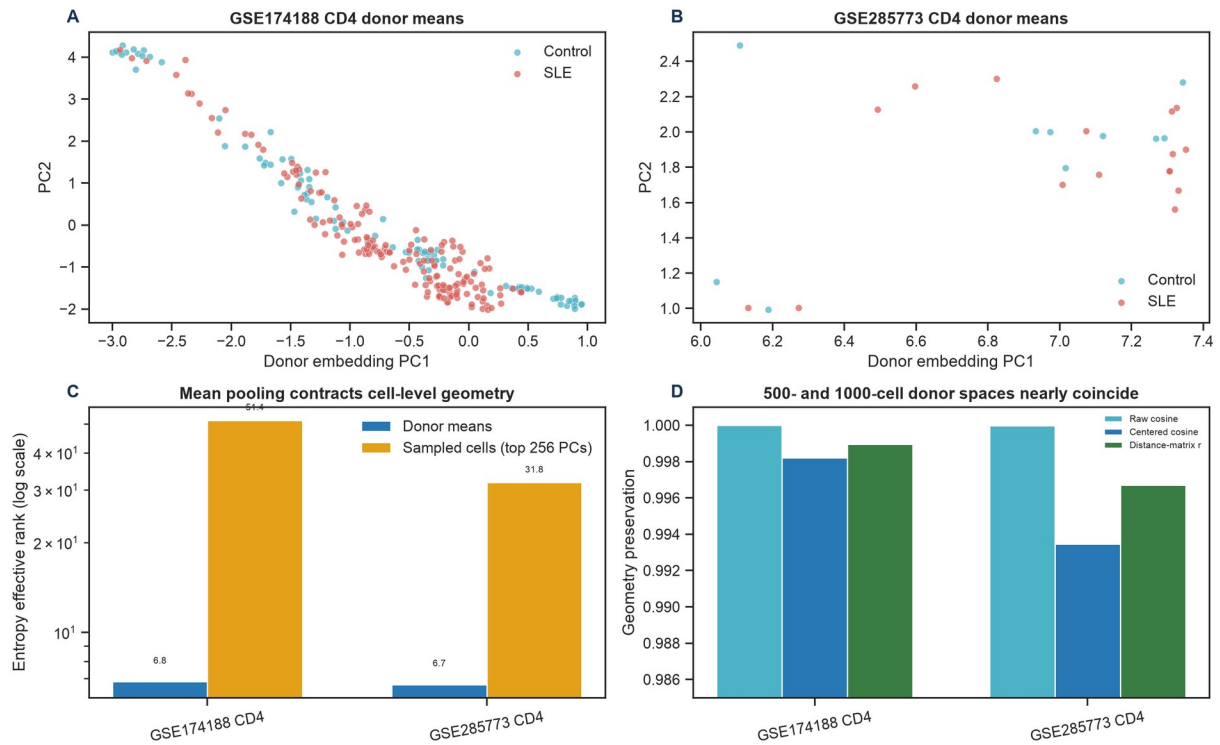
Supplementary Figure S28. Engineering and nuisance-variable probes. Panel A compares classifier and feature-scaling choices for fixed Geneformer donor embeddings. Panel B contrasts total-cell-count-only, Geneformer, and the auxiliary whole-PBMC cell-composition baseline. Panel C shows the incomplete model-size and sequence-length grid. Panel D compares three GSE285773 cell-sampling seeds with fixed donor folds. These panels are descriptive implementation sensitivities.

External cohort shift separates ranking performance from probability reliability



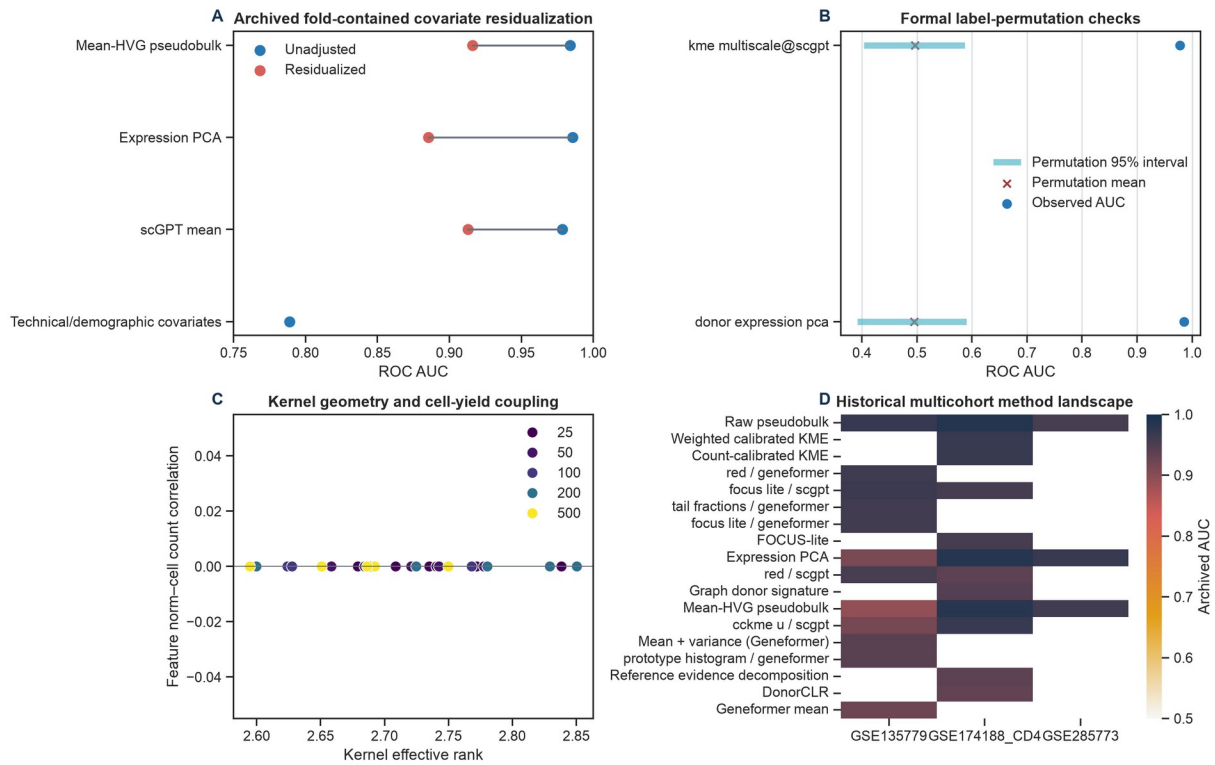
Supplementary Figure S29. Shared-fold transfer and calibration sensitivity. Panels A-B compare the strict source-only transfer estimates with a sensitivity in which all primary representations use the same source-internal donor folds for regularisation selection. Panel C shows paired target-donor probabilities in the larger target, and panel D contrasts discrimination with Brier score and 10-bin ECE. The larger-target pseudobulk ordering is preserved.

Frozen donor means occupy a low-dimensional, cohort-sensitive subspace

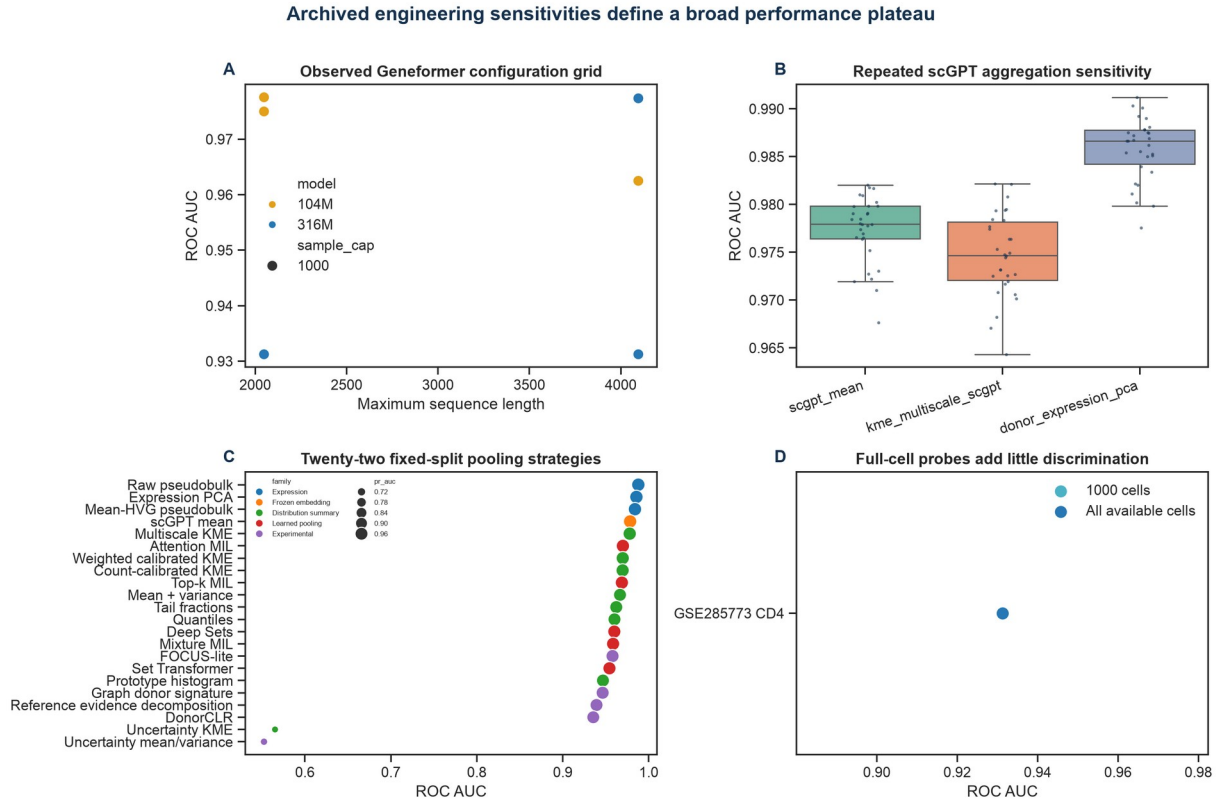


Supplementary Figure S30. Geometry of frozen donor-mean Geneformer embeddings. Panels A-B show label-blind donor PCA coordinates in the reciprocal-transfer cohorts. Panel C contrasts entropy effective rank at the donor and sampled-cell levels. Panel D quantifies cap500-cap1000 preservation of donor geometry. The analysis is descriptive and does not identify a causal mechanism for the transfer difference.

Validity controls reveal both biological signal and nuisance exposure



Supplementary Figure S31. Consolidated validity controls and historical method landscape. Panel A shows archived training-fold covariate residualization. Panel B shows accepted 1,000-permutation intervals. Panel C relates kernel effective rank to cell-yield coupling. Panel D assembles traceable historical method records. The archived branches are supporting sensitivities rather than matched primary comparisons.



Supplementary Figure S32. Consolidated resource and aggregation atlas. Panels A-B show model-configuration and repeated scGPT aggregation sensitivities. Panel C ranks 22 fixed-split donor representations. Panel D compares the Geneformer 1,000-cell cap with all available cells. These engineering sensitivities define a broad plateau but do not replace the reciprocal external-transfer estimand.

Data and code availability

The study uses public data from GSE135779, GSE174188, and GSE285773. The reproducibility package includes analysis scripts, donor-level predictions and metrics, fold assignments, compact input matrices required for the reported donor-level analyses, input and embedding checksums, matched random-module definitions, scGPT provenance, the final cap500 pooling tables and reproduction gate, the evidence ledger, and environment locks. The preserved 607,215,234-byte cap500 server archive has SHA256 5f08f97007f4358039afb85185816d298ce222420c0e221f2732d6d3c5f32c5d. A versioned Zenodo release corresponding to this manuscript will be deposited before publication; the release DOI will be inserted after the record is created.

Ethics statement

This study reanalysed public, de-identified data.

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